Fluid/Movable Antenna Meets Active RIS: A Next Generation ISAC System Design

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***Abstract*—This paper investigates a fluid antenna system (FAS)/movable antenna (MA) and active reconfigurable intelligent surface (RIS)-aided integrated sensing and communication (ISAC) system. Specifically, we maximize the radar signal-to-noise-ratio (SNR) by jointly optimizing the beamforming matrix at the base station (BS), the reflecting coefficients of active RIS and port selection of FAS, subject to the transmit power constraints and the communication requirements of users. To tackle the problem, we propose an effective algorithm to decompose the main problem into three subproblems. For each subproblem, we first utilize semidefinite relaxation (SDR), successive convex approximation (SCA) and penalty function to reformulate the subproblem, then the majorization-minimization (MM) method is applied to solve the subproblem iteratively. Numerical results validate the superiority of the proposed system. In addition, FAS can reduce the resource consumption and active RIS can overcome the severe path loss effectively.**

***Index Terms*—Fluid antenna system (FAS), reconfigurable intelligent surface (RIS), active RIS, integrated sensing and communication (ISAC).**

I. Introduction

IN THE era of the fifth-generation (5G) and earlier generations of wireless communication, the systems of communication and sensing only have limited intersection. However, this paradigm is expected to shift in future sixth-generation (6G) network and the emergence of integrated sensing and communication (ISAC) technique is considered as a convergence point, bridging the gap of two realms [1], [2]. As a key technique of 6G network, the concept of ISAC is not an imagination and the integration of communication and radar systems is reciprocal. By designing the signal processing module, hardware platform and resource allocation, ISAC can not only enhance spectrum efficiency, but also obtain the integration gain and coordination gain [1]. Due to these advantages, ISAC can be applied in various scenarios including gesture recognition, environmental imaging and smart home [1], [2], [3], [4], [5]. The ISAC paradigm can be categorized into two main approaches: radar-communication coexistence (RCC) and dual-functional radar and communication (DFRC) [2]. The former prioritizes the sharing of spectrum while maintaining separation for the hardware platform and the transmission signals [6] while the latter achieves deeper integration by sharing both hardware platform and spectrum [2]. Currently, the combination of DFRC systems and the new technologies of wireless communication, such as reconfigurable intelligent surface (RIS) and fluid antenna system (FAS)/movable antenna (MA), has spurred academic interests significantly. As one of the key candidate technologies for future 6G communications, RIS has attracted increasing attention due to its hardware architecture [7], [8], [9], [10], [11], [12]. RIS is a metamaterial-based planar structure which comprises various passive reflecting elements. Thus, RIS can enhance the desired signal by intelligently adjusting the phase of incident signal. Besides, if the user is located in the dead zone of BS, RIS can construct a virtual line-of-sight (LoS) link between BS and user. These characteristics of RIS enhance the overall performance and the coverage range of the signal [13], [14], [15], [16], [17], [18]. However, the signals reflected by conventional RIS suffer severe path loss. Recently, a novel architecture of RIS, termed as active RIS, overcomes the limitations of passive RIS by incorporating active amplifiers into each reflecting element [19], [20], [21], [22], [23], [24]. Unlike passive RIS, active RIS can not only adjust the phase of incident signal, but also amplify it, which mitigates the "multiplicative fading". Some researchers have demonstrated that the active RIS-aided systems achieve higher achievable rate with the same power budget and fewer elements [19], [20]. Besides, the overcome of the severe path loss enhances the communication and sensing performance significantly [25], [26], [27], [28], [29], [30].

Conventional MIMO systems usually rely on fixed position antenna (FPA) arrays, which limits their ability to utilize the spatial variation of channel characteristics. Recently, a new technique, known as FAS or MA,[1] has received widespread attention in academic society due to its ability of adjusting the geometric configuration of antenna and reconfiguring the channel [31], [32], [33], [34], [35], [36]. The unique feature of FAS enables it to overcome the degrees of

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[1]For conciseness, we use FAS to represent FAS/MA in the remainder of this paper.

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freedom (DoFs) limitation of FPA and improve the overall performance [33].

A. Related Works

1) Researches on Beamforming Design of DFRC Systems: In DFRC systems, allocating limited resources strategically to achieve the trade-off between communication and sensing performance has spurred research interests recently [37]. Some researches focus on the communication performance while ensuring sensing performance. Authors of [38] studied the physical layer security in an ISAC system and minimized the maximum eavesdropping signal-to-interference-plus-noise-ratio (SINR) while ensuring the SINR of the targets. In [39], the authors considered a full-duplex (FD) ISAC system and maximized the communication rate while maintaining accurate sensing performance. In contrast, other works prioritize the sensing performance. For example, the authors of [40] minimized the beampattern matching error while satisfying the SINR requirements of users. They also demonstrated that the dedicated radar signal is unnecessary for LoS communication channels. Some researches further considered the flexible trade-off between communication and sensing requirement. For instance, in [41], the authors proposed a series of optimal transmit beamforming design of a DFRC system to minimize the weighted multi-user interference and beampattern matching error. The weighted sensing mutual information (MI) and the communication total data rate were maximized in [42].

2) Researches on RIS-Aided Communication and Sensing: For ISAC systems, high-frequency signals can enable ultra high transmit rate and ultra-precise sensing. Nevertheless, their weak penetration capability renders them highly susceptible to blockage, degrading the communication and sensing performance significantly. Fortunately, RIS offers a prospective solution for the aforementioned blockage issue by manipulating the phase shift and constructing virtual LoS link. In [18], the authors considered only the reflected link and proposed a joint transmit and reflective beamforming design for a RIS-aided ISAC system to maximize the minimum beampattern gains. Furthermore, in [43], the authors considered both direct and reflected links and maximized the radar MI.

However, as mentioned previously, conventional RIS is composed of passive elements, which causes the signals to suffer severe path loss, known as the "multiplicative fading" effect. Specifically, the path loss of the cascaded link is the product of those in single links. Thus, deploying the active RIS is an effective approach to mitigate the path loss. For instance, in terms of sensing performance, the authors in [44] investigated the radar SINR maximization problem in an active RIS-aided DFRC system with four-hop link. Besides, the authors demonstrated that the deployment of active RIS can mitigate the "multiplicative fading" impact. The double active RIS-aided RCC system was studied in [45]. Furthermore, for a fundamental estimation theory perspective, the authors in [46] derived and minimized the Cramér-Rao bound (CRB) and validated the potential of active RIS to improve estimation performance. Some researchers studied the physical layer security in active RIS-aided ISAC systems. For example, the secure rate maximization problem was investigated in [47]. However, in the considered scenario in [47], the active RIS only reflects the communication signal. The authors of [48] considered the four-hop reflection and demonstrated the superiority of active RIS in enhancing the secure data rates and radar signal-to-noise-ratio (SNR). Nevertheless, the aforementioned ISAC systems all utilize FPA arrays.

3) Researches on FAS-Aided Communication and Sensing: Existing researches have demonstrated that the deployment of FAS can enhance the communication performance [49], [50], [51], [52], [53], [54]. Building upon the explorations in wireless communications, some researches have attempted to apply FAS in ISAC systems. For example, the authors of [55] and [56] enabled one-dimensional (1D) FAS at the transmitter (Tx) in ISAC systems. Furthermore, in [57] and [58], the authors deployed two-dimensional (2D) FAS in ISAC systems, and numerical results illustrated that FAS-aided systems achieve better overall performance than the systems aided by FPA. Moreover, the authors of [59] first proposed to apply FAS in a RIS-aided ISAC system. However, the investigation of deploying active RIS in an FAS-aided ISAC system remains unexplored.

While these studies highlight the growing application potential of FAS in ISAC, they often rely on an idealized assumption of continuous antenna positioning. Due to intrinsic hardware constraints of FAS, discrete position configurations represent a more practical paradigm [60], [61]. Recently, some researchers have considered discrete scenarios [62], [63], [64], [65], [66]. For instance, the authors of [62] transformed the discrete position optimization into the binary programming and utilized the generalized Bender's decomposition (GBD) to tackle it. However, the worst-case complexity of GBD increases exponentially as the number of antennas grows. Moreover, the authors of [63] investigated port selection of FAS in an ISAC system. Nevertheless, the authors considered only 1D scenario.

B. Our Contributions

Based on the above discussion, this paper considers an FAS and active RIS-aided ISAC system. The differences between related works and our work were listed in Table I. Our main contributions are summarized as follows:

1) We consider a FAS and active RIS-aided ISAC system, where 2D FAS is enabled at the DFRC Tx to fully utilize the spatial DoFs. Since the direct link between the Tx and the radar target is obstructed, establishing a virtual LoS link becomes necessary. Thus, the deployment of the active RIS can improve the channel characteristics and enhance both communication and sensing performance. Then, we formulate the maximization problem of sensing SNR by jointly optimizing the beamforming matrix, reflecting coefficients of active RIS and port selection of FAS, subject to the communication SINR and transmit power constraints.
2) In order to tackle the intrinsic non-convexity of the optimization problem, we propose an alternating optimization (AO)-based algorithm incorporates

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TABLE I
COMPARISON BETWEEN RELATED WORKS AND OUR WORK

	FAS	RIS	Optimization objective	Optimization methods
[18]	×	Passive	Minimum beampattern gain maximization	SDR
[43]	×	Passive	Radar MI maximization	SDR and one-dimension iterative (ODI)
[44]	×	Active	Radar SINR maximization	MM, SDR and SCA
[45]	×	Active	Achievable data rate maximization	Penalty dual decomposition (PDD) and block coordinate descent (BCD)
[46]	×	Active	CRB for target direct-of-arrival (DoA) estimation minimization	MM and SDR
[47]	×	Active	Communication secrecy rate maximization	MM, SDR and SCA
[48]	×	Active	Communication secrecy rate maximization	MM, SDR and SCA
[55]	1D, continuous position	×	Communication sum rate maximization	MM, proximal distance algorithm (PDA) and extrapolated projected gradient (EPG)
[56]	1D, continuous position	×	Weighted communication sum rate and sensing MI maximization	fractional programming (FP) and search-based projected gradient ascent (SPGA)
[57]	2D, continuous position	×	Sensing signal-clutter-noise ratio (SCNR) maximization	MM
[58]	2D, continuous position	×	Radar SINR maximization	Charnes-Cooper transformation, SCA and SDR
[59]	2D, continuous position	Passive	Minimum beampattern gain maximization	SDR, sequential rank-one constraint relaxation (SRCR) and SCA
[63]	1D, discrete position	×	Communication sum rate maximization	SCA, SDR and penalty approach
Proposed	2D, discrete position	Active	Radar SNR maximization	MM, SDR, SCA and penalty approach

majorization-minimization (MM), semidefinite relaxation (SDR) and successive convex approximation (SCA) methods. Specifically, we optimize the beamforming matrix by utilizing the SDR method and reflecting coefficients of active RIS by applying the MM method. In terms of the ports of fluid antennas, we first transform the discrete optimization into binary integer programming, then adopt the SCA methods and penalty function to tackle it.

3) Numerical results illustrate the advantages of the FAS and active RIS-aided ISAC system. Meanwhile, the implementation of FAS can reduce the resource consumption and the deployment of active RIS can effectively overcome the severe path loss.

C. Organizations

This paper is organized as follows: Section II illustrates the system model of the FAS and active RIS-aided ISAC system and formulates the optimization problem. In Section III, we propose the AO-based algorithm. Section IV presents the numerical results and Section V concludes this paper.

D. Notations

The italics letter x or X, the boldface lowercase letter $\mathbf{x}$ and the boldface uppercase letter $\mathbf{X}$ represent values, vectors and matrices, respectively. $(\cdot)^*$, $(\cdot)^T$ and $(\cdot)^H$ denote the conjugate, transpose and Hermitian operators, respectively. $\mathbb{C}^{M\times N}$ and $\mathbb{R}^{M\times N}$ represent the set of $M \times N$ complex and real matrices. $\mathcal{CN}(\mathbf{m}, \mathbf{C})$ denotes that a random variable follows the complex Gaussian distribution with a mean vector of $\mathbf{m}$ and a covariance matrix of $\mathbf{C}$. $\mathbb{E}[\cdot]$ denotes the expectation operation. For a complex value a, $|a|$ denotes the modulus of a. Besides, $\mathrm{Re}(a)$ and $\mathrm{Im}(a)$ denote the real and imaginary part of a, respectively. $\mathrm{Tr}(\mathbf{X})$ and $\mathrm{rank}(\mathbf{X})$ denote the trace and the rank of $\mathbf{X}$, respectively. $\mathbf{X} \succeq 0$ illustrates that $\mathbf{X}$ is a positive semidefinite matrix. $\mathbf{X}^{-1}$ represents the inverse of $\mathbf{X}$. $\|\mathbf{x}\|_2$ denotes the 2-norm of $\mathbf{x}$. $\mathbf{A} \odot \mathbf{B}$ and $\mathbf{A} \otimes \mathbf{B}$ denote the Hadamard product and Kronecker product of $\mathbf{A}$ and $\mathbf{B}$, respectively. $\mathrm{diag}(\mathbf{X})$ represents a vector whose elements are the main diagonal elements of $\mathbf{X}$, and $\mathrm{diag}(\mathbf{x})$ represents a diagonal matrix containing the elements of $\mathbf{x}$ on its main diagonal. $\mathrm{vec}(\mathbf{X})$ represents the vectorization of $\mathbf{X}$. The element in the i-th row and j-th column of $\mathbf{X}$ is denoted as $[\mathbf{X}]_{i,j}$.

II. SYSTEM MODEL AND PROBLEM FORMULATION

As shown in Fig. 1, we consider an ISAC system aided by FAS and active RIS, which comprises a radar target, communication users, a DFRC Tx and a DFRC receiver (Rx). Specifically, the Tx equipped with M fluid antennas serves K users and each user is equipped with a single antenna. Besides, the active RIS

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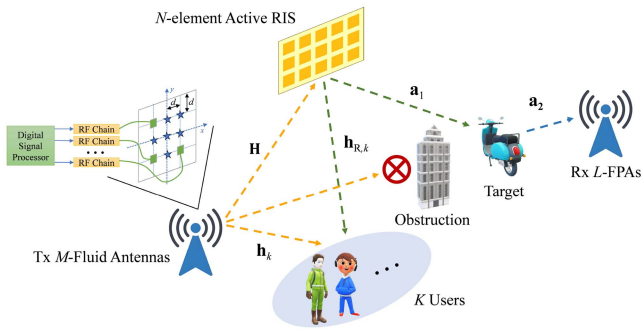

Fig. 1. FAS and active RIS-aided ISAC system.

with N elements is deployed to establish the virtual LoS link between Tx and target. Moreover, the Rx equipped with L FPAs is deployed to receive the echo signal.

A. Signal Model

The transmit signal at Tx is formulated as

$$\mathbf{x} = \mathbf{W}_r\mathbf{s}_r + \mathbf{W}_c\mathbf{s}_c = \left[\mathbf{W}_r, \mathbf{W}_c\right]\left[{\mathbf{s}_r}^T, {\mathbf{s}_c}^T\right]^T = \mathbf{W}\mathbf{s}, \quad (1)$$

where $\mathbf{s} \in \mathbb{C}^{M+K}$ denotes the transmit symbols with $\mathbf{s}_r \in \mathbb{C}^M$ and $\mathbf{s}_c \in \mathbb{C}^K$ representing the radar signal and communication symbols, respectively. Moreover, the radar signal and communication symbols are assumed to be uncorrelated and the transmit symbols follow the distribution of $\mathcal{CN}(\mathbf{0}, \mathbf{I}_{M+K})$. Besides, $\mathbf{W} \in \mathbb{C}^{M\times(M+K)}$ represents the transmit beamforming matrix with $\mathbf{W}_r \in \mathbb{C}^{M\times M}$ and $\mathbf{W}_c \in \mathbb{C}^{M\times K}$ denoting the bramforming matrices for radar signal and communication symbols, respectively. Thus, the covariance matrix of $\mathbf{x}$ is given by

$$\mathbf{R} = \mathbb{E}\left[\mathbf{x}\mathbf{x}^H\right] = \mathbf{W}\mathbf{W}^H = \mathbf{W}_r\mathbf{W}_r^H + \sum_{k=1}^{K}\mathbf{R}_k, \quad (2)$$

where $\mathbf{R}_k \triangleq \mathbf{w}_{c,k}\mathbf{w}_{c,k}^H \in \mathbb{C}^{M\times M}$ and $\mathbf{w}_{c,k}$ is the k-th column of $\mathbf{W}_c$.

B. Channel Model

We assume that the moving region of FAS is negligible compared to the signal propagation distance, satisfying the far-field condition. Thus, the planar wave model is adopted for the Tx-User and Tx-RIS links [49]. Moreover, we assume that only the phases of multi-path channels vary with fluid antenna's movement, while the amplitudes, the angle of arrivals (AoAs) and the angle of departures (AoDs) remain constant.

The FAS-aided system is illustrated in Fig. 1, and the pentagrams represent all available ports. Then, we assume that the fluid antennas move within a 2D square region.

Besides, we denote the Q available ports set as $\mathcal{P} \triangleq \{\mathbf{p}_1, \cdots \mathbf{p}_Q\}$, where $\mathbf{p}_q = [x_q^P, y_q^P]^T, 1 \le q \le Q$ represents the coordinates of the q-th position of ports. Next, the distance between adjacent available ports is d in both x and y directions. Then, the m-th activated port is denoted as $\mathbf{t}_m = [x_m, y_m]^T \in \mathcal{P}, 1 \le m \le M$. For the Tx-RIS link, $\theta_i^t \in [0, \pi]$ and $\phi_i^t \in [0, \pi], 1 \le i \le L_R^t$ are the elevation and azimuth angles of the i-th transmit path, respectively. Similarly, $\theta_j^r \in [0, \pi]$ and $\phi_j^r \in [0, \pi], 1 \le j \le L_R^r$ represent those for the j-th receive path. Here, L_R^t and L_R^r are the numbers of transmit and receive paths, respectively. Similarly, those for the Tx-User link are defined as $L_U^{t,k}$ and $L_U^{r,k}$, respectively. Next, the transmit field response vector (FRV) for the m-th fluid antenna in the Tx-RIS link is formulated as

$$\mathbf{g}(\mathbf{t}_m) = \left[e^{j\frac{2\pi}{\lambda}\rho_1^t(\mathbf{t}_m)}, \cdots, e^{j\frac{2\pi}{\lambda}\rho_{L_R}^t(\mathbf{t}_m)}\right]^T \in \mathbb{C}^{L_R^t}, \quad (3)$$

where λ denotes the carrier wavelength and $\rho_i^t(\mathbf{t}_m) = (x_m - x_1)\sin\theta_i^t\cos\phi_i^t + (y_m - y_1)\cos\theta_i^t$ represents the difference of the signal propagation distance between $\mathbf{t}_m$ and $\mathbf{t}_1$ for the i-th channel path. Then, the transmit field response matrix (FRM) can be expressed as

$$\mathbf{G}(\mathbf{t}) = [\mathbf{g}(\mathbf{t}_1), \cdots, \mathbf{g}(\mathbf{t}_m)] \in \mathbb{C}^{L_R^t\times M}, \quad (4)$$

where $\mathbf{t} \triangleq \{\mathbf{t}_m\}_{m=1}^M$. Similarly, the FRM for the Tx-User link is given by

$$\mathbf{G}_k(\mathbf{t}) = [\mathbf{g}_k(\mathbf{t}_1), \cdots, \mathbf{g}_k(\mathbf{t}_m)] \in \mathbb{C}^{L_U^{t,k}\times M}, 1 \le k \le K. \quad (5)$$

In addition, the receive FRV and FRM of the n-th reflection element for the Tx-RIS link are given by

$$\mathbf{f}(\mathbf{r}_n) = [e^{j\frac{2\pi}{\lambda}\rho_1^r(\mathbf{r}_n)}, \cdots, e^{j\frac{2\pi}{\lambda}\rho_{L_R}^r(\mathbf{r}_n)}]^T \in \mathbb{C}^{L_R^r}, \quad (6)$$

$$\mathbf{F}(\mathbf{r}) = [\mathbf{f}(\mathbf{r}_1), \cdots, \mathbf{f}(\mathbf{r}_N)] \in \mathbb{C}^{L_R^r\times N}, \quad (7)$$

where $\mathbf{r}_n$ is the coordinate for the n-th reflection element of active RIS with $\mathbf{r} \triangleq \{\mathbf{r}_n\}_{n=1}^N$. Then, the path response matrices (PRMs) $\mathbf{\Sigma} \in \mathbb{C}^{L_R^r\times L_R^t}$ and $\mathbf{\Sigma}_k \in \mathbb{C}^{L_U^{r,k}\times L_U^{t,k}}, 1 \le k \le K$ represent the responses of all paths in the Tx-RIS/User link, respectively. Subsequently, the channel matrix of the Tx-RIS link is expressed as

$$\mathbf{H}(\mathbf{t}) = \mathbf{F}(\mathbf{r})^H\mathbf{\Sigma}\mathbf{G}(\mathbf{t}) \in \mathbb{C}^{N\times M}. \quad (8)$$

Similarly, the channel vectors of the Tx-User link are

$$\mathbf{h}_k^H(\mathbf{t}) = \mathbf{1}^H\mathbf{\Sigma}_k\mathbf{G}_k(\mathbf{t}) \in \mathbb{C}^{1\times M}. \quad (9)$$

Moreover, since the positions of RIS elements and the antennas of users are fixed, we denote the constant channel vectors of the RIS-User link by $\mathbf{h}_{\mathrm{R},k}^H \in \mathbb{C}^{1\times N}$. Besides, we assume that the perfect channel state information (CSI) is available through effective channel estimation.

For the radar target, the multi-reflected signals may experience severe path loss, hence the effect of the non-line-of-sight (NLoS) channel in the RIS-Target link and Target-Rx link can be neglected, and only the LoS path can be successfully detected here. Let θ_1 and θ_2 represent the AoD of the target related to the active RIS and the AoA of the Rx related to the target, respectively. Then, the steer vectors at RIS and Rx are given by

$$\mathbf{a}_1(\theta_1) = \left[1, e^{j\frac{2\pi d_{\mathrm{RIS}}}{\lambda}\sin\theta_1}, \cdots, e^{j\frac{2\pi d_{\mathrm{RIS}}}{\lambda}(N-1)\sin\theta_1}\right]^T, \quad (10)$$

$$\mathbf{a}_2(\theta_2) = \left[1, e^{j\frac{2\pi d_{\mathrm{Rx}}}{\lambda}\sin\theta_2}, \cdots, e^{j\frac{2\pi d_{\mathrm{Rx}}}{\lambda}(L-1)\sin\theta_2}\right]^T, \quad (11)$$

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where d_{RIS} and d_{Rx} are the RIS element spacing and Rx antenna spacing, respectively. Hence, the channel matrix in the RIS-Target-Rx link can be expressed as

$$\mathbf{A} = \beta_p \mathbf{a}_2(\theta_2)\mathbf{a}_1^H(\theta_1) \in \mathbb{C}^{L\times N}, \tag{12}$$

where $\beta_p \in \mathbb{C}$ is the complex path loss in the RIS-Target-Rx link.

C. Active RIS Model

For an active RIS system, the reflecting coefficients matrix is given by $\mathbf{\Psi} = \text{diag}(\psi_1, \cdots, \psi_N)$, where $\psi_n = \alpha_n e^{j\varphi_n}, 1 \leq n \leq N$. Besides, α_n and φ_n are amplitude and phase shift, respectively. Since the hardware conditions of RIS, α_n should be less than the maximum amplification factor $\alpha_{\max}$. Next, the reflected signal is

$$\mathbf{y}_1 = \mathbf{\Psi}\left(\mathbf{H}\mathbf{x} + \mathbf{n}_{\text{R}}\right) \in \mathbb{C}^N, \tag{13}$$

where $\mathbf{n}_{\text{R}} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{\sigma}_{\text{R}}^2 \mathbf{I}_N)$ represents the additive white Gaussian noises (AWGN) at the active RIS.

D. Communication and Radar Performance Metrics

According to the previous derivations, the received signal of the k-th user is

$$\begin{aligned} y_k &= \mathbf{h}_k^H \mathbf{x} + \mathbf{h}_{\text{R},k}^H \mathbf{\Psi}(\mathbf{H}\mathbf{x} + \mathbf{n}_{\text{R}}) + n_k \\ &= (\mathbf{h}_k^H + \mathbf{h}_{\text{R},k}^H \mathbf{\Psi}\mathbf{H})\mathbf{x} + \mathbf{h}_{\text{R},k}^H \mathbf{\Psi}\mathbf{n}_{\text{R}} + n_k, \end{aligned} \tag{14}$$

where $n_k \sim \mathcal{CN}(0, \sigma_k^2)$ is the AWGN at the k-th user. Subsequently, the SINR for the k-th user is

$$\begin{aligned} \gamma_k &= \frac{\mathbf{h}_{e,k}^H \mathbf{w}_{c,k}\mathbf{w}_{c,k}^H \mathbf{h}_{e,k}}{\sum_{i=1, i\neq M+k}^{M+K} \mathbf{h}_{e,k}^H \mathbf{w}_i \mathbf{w}_i^H \mathbf{h}_{e,k} + \sigma_{\text{R}}^2 \mathbf{h}_{\text{R},k}^H \mathbf{\Psi}\mathbf{\Psi}^H \mathbf{h}_{\text{R},k} + \sigma_k^2} \\ &= \frac{\mathbf{h}_{e,k}^H \mathbf{R}_k \mathbf{h}_{e,k}}{\sum_{i=1, i\neq k}^{K} \mathbf{h}_{e,k}^H \mathbf{R}_i \mathbf{h}_{e,k} + \sigma_{\text{R}}^2 \mathbf{h}_{\text{R},k}^H \mathbf{\Psi}\mathbf{\Psi}^H \mathbf{h}_{\text{R},k} + \sigma_k^2} \\ &= \frac{\mathbf{h}_{e,k}^H \mathbf{R}_k \mathbf{h}_{e,k}}{\mathbf{h}_{e,k}^H (\mathbf{R} - \mathbf{R}_k)\mathbf{h}_{e,k} + \sigma_{\text{R}}^2 \mathbf{h}_{\text{R},k}^H \mathbf{\Psi}\mathbf{\Psi}^H \mathbf{h}_{\text{R},k} + \sigma_k^2}, \end{aligned} \tag{15}$$

where $\mathbf{h}_{e,k}^H \triangleq \mathbf{h}_k^H + \mathbf{h}_{\text{R},k}^H \mathbf{\Psi}\mathbf{H}$ is the equivalent channel of the Tx-User link.

Moreover, the echo signal at the Rx is given by

$$\mathbf{y}_r = \mathbf{A}\mathbf{\Psi}(\mathbf{H}\mathbf{x} + \mathbf{n}_{\text{R}}) + \mathbf{n}_r \in \mathbb{C}^L, \tag{16}$$

where $\mathbf{n}_r \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{\sigma}_r^2 \mathbf{I}_L)$ denotes the AWGN at the Rx. Thus, the radar SNR can be formulated as

$$\gamma_r = \text{Tr}\left(\mathbf{A}\mathbf{\Psi}\mathbf{H}\mathbf{R}\mathbf{H}^H \mathbf{\Psi}^H \mathbf{A}^H \mathbf{J}^{-1}\right), \tag{17}$$

where $\mathbf{J} \triangleq \sigma_{\text{R}}^2 \mathbf{A}\mathbf{\Psi}\mathbf{\Psi}^H \mathbf{A}^H + \sigma_r^2 \mathbf{I}_L$.

E. Problem Formulation

In this paper, we aim to maximize the radar SNR γ_r by jointly optimizing the covariance matrix $\mathbf{R}$ of the transmit beamforming matrix, the reflecting coefficients of the active RIS and the port selection of FAS while satisfying the constraints. Subsequently, the optimization problem is formulated as

$$\begin{aligned} \max_{\mathbf{R},\mathbf{\Psi},\mathbf{t}} \quad & \gamma_r && \text{(18a)} \\ \text{s.t.} \quad & \gamma_k \geq \gamma_{th}, \forall k, && \text{(18b)} \\ & \text{Tr}(\mathbf{R}) \leq P_0, && \text{(18c)} \\ & \mathbf{R} \succeq 0, \mathbf{R}_k \succeq 0, \forall k, && \text{(18d)} \\ & \text{rank}(\mathbf{R}_k) = 1, \forall k, && \text{(18e)} \\ & \mathbf{\Psi} = \text{diag}(\psi_1, \cdots, \psi_N), && \text{(18f)} \\ & \alpha_n \leq \alpha_{\max}, 1 \leq n \leq N, && \text{(18g)} \\ & \mathbb{E}\left[\mathbf{y}_1^H \mathbf{y}_1\right] \leq P_1, && \text{(18h)} \\ & \|\mathbf{t}_m - \mathbf{t}_q\|_2 \geq D_{\min}, m \neq q, && \text{(18i)} \\ & \mathbf{t}_m \in \mathcal{P}, 1 \leq m \leq M, && \text{(18j)} \end{aligned}$$

where γ_{th} in (18b) denotes the SINR requirement for communication users. P_0 and P_1 in (18c) and (18h) denote the maximum transmit power of the Tx and the RIS, respectively. The constraint (18i) ensures that the distance between every two fluid antennas is greater than or equal to a constant $D_{\min} \geq 0$ to avoid the coupling effect of the antennas.

We notice that the variables $\mathbf{R}$, $\mathbf{\Psi}$ and $\mathbf{t}$ are coupled intricately, which poses significant challenges to Problem (18). In addition, the non-convex constraints (18b) (18e) (18i) (18j) further complicate the problem.

III. Proposed Algorithm

In this section, in order to tackle the aforementioned challenges, we employ the AO method to decouple the problem into three subproblems and solve the subproblems alternatively. Specifically, we first utilize the SDR method to optimize the covariance matrix $\mathbf{R}$. Then, the reflecting coefficients of the active RIS $\mathbf{\Psi}$ are optimized by the MM method. For the port selection of FAS, we first transform the discrete constraint into the binary selection matrix. Then, we apply the SCA method to relax the binary constraint. Finally, we incorporate the relaxed binary inequality into the objective function as a penalty term.

A. Optimize $\mathbf{R}$

In this subsection, we tackle the optimization of the covariance matrix $\mathbf{R}$ with given $\mathbf{t}$ and $\mathbf{\Psi}$.

1) Active RIS Transmission Power Constraint: The constraint (18h) is reformulated as

$$\text{Tr}\left(\mathbf{B}_1 \mathbf{R}\right) \leq b_p, \tag{19}$$

where $b_p \triangleq P_1 - \boldsymbol{\sigma}_{\text{R}}^2 \text{Tr}(\mathbf{\Psi}\mathbf{\Psi}^H)$ and $\mathbf{B}_1 \triangleq \mathbf{H}^H \mathbf{\Psi}^H \mathbf{\Psi}\mathbf{H}$.

2) Communication SINR Constraint: The constraint of the required SINR for all communication users (18b) can be transformed as

$$\left(1 + \gamma_{th}^{-1}\right) \text{Tr}\left(\mathbf{H}_{e,k}\mathbf{R}_k\right) \geq \text{Tr}\left(\mathbf{H}_{e,k}\mathbf{R}\right) + b_k, 1 \leq k \leq K, \tag{20}$$

where $b_k \triangleq \sigma_{\text{R}}^2 \mathbf{h}_{\text{R},k}^H \mathbf{\Psi}\mathbf{\Psi}^H \mathbf{h}_{\text{R},k} + \sigma_k^2$ and $\mathbf{H}_{e,k} \triangleq \mathbf{h}_{e,k}\mathbf{h}_{e,k}^H$.

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3) Problem Reformulation: Based on the previous derivations, the covariance matrix optimization problem is recast as

$$\max_{\mathbf{R},\{\mathbf{R}_k\}} \quad \mathrm{Tr}\left(\mathbf{B}_2\mathbf{R}\right) \tag{21a}$$

$$\text{s.t.} \quad \left(1+\gamma_{th}^{-1}\right)\mathrm{Tr}\left(\mathbf{H}_{e,k}\mathbf{R}_k\right) \geq \mathrm{Tr}\left(\mathbf{H}_{e,k}\mathbf{R}\right)+b_k, \forall k, \tag{21b}$$

$$\mathrm{Tr}(\mathbf{R}) \leq P_0, \tag{21c}$$

$$\mathrm{Tr}\left(\mathbf{B}_1\mathbf{R}\right) \leq b_p, \tag{21d}$$

$$\mathbf{R} \succeq 0, \mathbf{R}_k \succeq 0, \forall k, \tag{21e}$$

$$\mathbf{R}-\sum_{k=1}^{K}\mathbf{R}_k \succeq 0, \tag{21f}$$

$$\mathrm{rank}(\mathbf{R}_k)=1, \forall k, \tag{21g}$$

where $\mathbf{B}_2 \triangleq \mathbf{H}^H\mathbf{\Psi}^H\mathbf{A}^H\mathbf{J}^{-1}\mathbf{A}\mathbf{\Psi}\mathbf{H}$. Problem (21) remains non-convex owing to the rank-one constraint (21g). Then the SDR method is adopted to handle the challenge. By relaxing constraint (21g), Problem (21) can be reformulated as

$$\max_{\mathbf{R},\{\mathbf{R}_k\}} \quad \mathrm{Tr}\left(\mathbf{B}_2\mathbf{R}\right) \tag{22a}$$

$$\text{s.t.} \quad \left(1+\gamma_{th}^{-1}\right)\mathrm{Tr}\left(\mathbf{H}_{e,k}\mathbf{R}_k\right) \geq \mathrm{Tr}\left(\mathbf{H}_{e,k}\mathbf{R}\right)+b_k, \forall k, \tag{22b}$$

$$\mathrm{Tr}(\mathbf{R}) \leq P_0, \tag{22c}$$

$$\mathrm{Tr}\left(\mathbf{B}_1\mathbf{R}\right) \leq b_p, \tag{22d}$$

$$\mathbf{R} \succeq 0, \mathbf{R}_k \succeq 0, \forall k, \tag{22e}$$

$$\mathbf{R}-\sum_{k=1}^{K}\mathbf{R}_k \succeq 0. \tag{22f}$$

Problem (22) is convex and can be solved by using CVX [67]. Subsequently, we introduce the following lemma to obtain a rank-one solution to Problem (21).

Lemma 1: There exists an optimal solution to Problem (21), which is denoted by $\hat{\mathbf{R}}, \{\hat{\mathbf{R}}_k\}$, satisfying

$$\mathrm{rank}\left(\hat{\mathbf{R}}_k\right)=1, \forall k. \tag{23}$$

Let $\tilde{\mathbf{R}}, \{\tilde{\mathbf{R}}_k\}$ denote the solution to Problem (22), the rank-one solution to Problem (21) $\hat{\mathbf{R}}, \{\hat{\mathbf{R}}_k\}$ can be expressed as

$$\hat{\mathbf{w}}_{c,k}=\left(\mathbf{h}_{e,k}^H\tilde{\mathbf{R}}_k\mathbf{h}_{e,k}\right)^{\frac{1}{2}}\tilde{\mathbf{R}}_k\mathbf{h}_{e,k}, \hat{\mathbf{R}}_k=\hat{\mathbf{w}}_{c,k}\hat{\mathbf{w}}_{c,k}^H, \forall k, \hat{\mathbf{R}}=\tilde{\mathbf{R}}. \tag{24}$$

Proof: Please refer to [68].

B. Optimize $\mathbf{\Psi}$

In this subsection, we optimize the reflecting coefficients of the active RIS $\mathbf{\Psi}$ without changing the value of $\mathbf{R}$ and $\mathbf{t}$.

1) Radar SNR Metric: To overcome the non-concave objective function γ_r, we design a lower bound of it by the first-order Taylor expansion shown in *Lemma 2*.

Lemma 2: Given $\mathbf{L}_{(t)}$ and $\mathbf{J}_{(t)}$ as the t-th iteration value of $\mathbf{L}$ and $\mathbf{J}$, we have

$$\begin{aligned}\mathrm{Tr}\left(\mathbf{L}^H\mathbf{J}^{-1}\mathbf{L}\right) &\geq 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{L}_{(t)}^H\mathbf{J}_{(t)}^{-1}\mathbf{L}\right)\right\} \\ &\quad -\mathrm{Tr}\left(\mathbf{J}_{(t)}^{-1}\mathbf{L}_{(t)}\mathbf{L}_{(t)}^H\mathbf{J}_{(t)}^{-1}\mathbf{J}\right).\end{aligned} \tag{25}$$

Proof: Please refer to Appendix A.

By leveraging *Lemma 2*, γ_r can be approximate as (26) which is shown at the bottom of this page. The parameters in (26) are defined as follows

$$\mathbf{C}_1 \triangleq \mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}_{(t)}^H\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\mathbf{A},$$

$$\mathbf{C}_2 \triangleq \sigma_{\mathrm{R}}^2\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\mathbf{A}\mathbf{\Psi}_{(t)}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}_{(t)}^H\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\mathbf{A},$$

$$c_r \triangleq \sigma_r^2\mathrm{Tr}\left(\mathbf{J}_{(t)}^{-1}\mathbf{A}\mathbf{\Psi}_{(t)}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}_{(t)}^H\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\right).$$

Then, we invoke the follow lemma to handle $\gamma_{r,lb}$.

Lemma 3: Let $\mathbf{U}_1 \in \mathbb{C}^{n\times n}$ and $\mathbf{U}_2 \in \mathbb{C}^{n\times n}$ be square matrices. Assuming that $\mathbf{M}=\mathrm{diag}(m_1,\cdots m_n)$ is a diagonal matrix, $\mathbf{m}=\mathrm{diag}(\mathbf{M})$ is a column vector, then we have

$$\mathrm{Tr}\left(\mathbf{M}^H\mathbf{U}_1\mathbf{M}\mathbf{U}_2\right)=\mathbf{m}^H\left(\mathbf{U}_1\odot\mathbf{U}_2^T\right)\mathbf{m}. \tag{27}$$

Assuming that $\mathbf{u}_2=\mathrm{diag}(\mathbf{U}_2)$, we have

$$\mathrm{Tr}\left(\mathbf{M}\mathbf{U}_2\right)=\mathbf{m}^T\mathbf{u}_2. \tag{28}$$

Proof: Please refer to [69].

By defining $\mathbf{v}=\mathrm{diag}(\mathbf{\Psi})$, we can turn $\gamma_{r,lb}$ into

$$\gamma_{r,lb}=2\mathrm{Re}\left\{\mathbf{c}_1^T\mathbf{v}\right\}-\mathbf{v}^H\left(\mathbf{C}_2\odot\mathbf{I}_N\right)\mathbf{v}-c_r, \tag{29}$$

where $\mathbf{c}_1=\mathrm{diag}(\mathbf{C}_1)$.

$$\begin{aligned}\gamma_r &= \mathrm{Tr}(\mathbf{A}\mathbf{\Psi}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}^H\mathbf{A}^H\mathbf{J}^{-1}) \\ &\geq 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{A}\mathbf{\Psi}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}_{(t)}^H\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\right)\right\}-\mathrm{Tr}\left(\mathbf{J}_{(t)}^{-1}\mathbf{A}\mathbf{\Psi}_{(t)}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}_{(t)}^H\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\mathbf{J}\right) \\ &= 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{\Psi}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}_{(t)}^H\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\mathbf{A}\right)\right\}-\mathrm{Tr}\left(\mathbf{\Psi}^H\sigma_{\mathrm{R}}^2\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\mathbf{A}\mathbf{\Psi}_{(t)}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}_{(t)}^H\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\mathbf{A}\mathbf{\Psi}\right) \\ &\quad -\sigma_r^2\mathrm{Tr}\left(\mathbf{J}_{(t)}^{-1}\mathbf{A}\mathbf{\Psi}_{(t)}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}_{(t)}^H\mathbf{A}^H\mathbf{J}_{(t)}^{-1}\right) \\ &\triangleq 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{\Psi}\mathbf{C}_1\right)\right\}-\mathrm{Tr}\left(\mathbf{\Psi}^H\mathbf{C}_2\mathbf{\Psi}\right)-c_r \triangleq \gamma_{r,lb}.\end{aligned} \tag{26}$$

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2) Communication SINR Constraint: The constraint of the required SINR for all communication users (18b) is rewritten as (30) shown at the bottom of this page, and we mark the left hand side of the second inequality as f. The further transformation of f, detailed in (31), is shown at the bottom of this page.

The parameters in (31) are defined as follows

$$\begin{aligned}\mathbf{C}_3 &\triangleq \mathbf{H}\left[\mathbf{R}-\left(1+\gamma_{th}^{-1}\right)\mathbf{R}_k\right]\mathbf{h}_k\mathbf{h}_{\mathrm{R},k}^H,\\ \mathbf{C}_{4,1} &\triangleq \mathbf{H}\left(\mathbf{R}-\mathbf{R}_k\right)\mathbf{H}^H+\sigma_{\mathrm{R}}^2\mathbf{I}_N,\\ \mathbf{C}_{4,2} &\triangleq -\gamma_{\mathrm{th}}^{-1}\mathbf{H}\mathbf{R}_k\mathbf{H}^H,\\ c_k &\triangleq \mathbf{h}_k^H\left[\mathbf{R}-\left(1+\gamma_{th}^{-1}\right)\mathbf{R}_k\right]\mathbf{h}_k+\sigma_k^2,\\ \mathbf{H}_{\mathrm{R},k} &\triangleq \mathbf{h}_{\mathrm{R},k}\mathbf{h}_{\mathrm{R},k}^H.\end{aligned}$$

Based on *Lemma 3*, we can turn (31) into

$$2\mathrm{Re}\left\{\mathbf{c}_3^T\mathbf{v}\right\}+\mathbf{v}^H\left[\mathbf{H}_{\mathrm{R},k}\odot\left(\mathbf{C}_{4,1}+\mathbf{C}_{4,2}\right)^T\right]\mathbf{v}+c_k\leq 0,\forall k, \tag{32}$$

where $\mathbf{c}_3=\mathrm{diag}(\mathbf{C}_3)$. We notice that $\mathbf{C}_{4,1}$ is a positive semidefinite matrix, whereas $\mathbf{C}_{4,2}$ is a negative semidefinite matrix. Therefore, we substitute $\mathbf{v}^H(\mathbf{H}_{\mathrm{R},k}\odot\mathbf{C}_{4,2}{}^T)\mathbf{v}$ with its first-order Taylor expansion of $\mathbf{v}$, that is

$$\begin{aligned}&\mathbf{v}^H\left(\mathbf{H}_{\mathrm{R},k}\odot\mathbf{C}_{4,2}^T\right)\mathbf{v}\\ &\leq 2\mathrm{Re}\left\{\mathbf{v}_{(t)}^H\left(\mathbf{H}_{\mathrm{R},k}\odot\mathbf{C}_{4,2}^T\right)\mathbf{v}\right\}-\mathbf{v}_{(t)}^H\left(\mathbf{H}_{\mathrm{R},k}\odot\mathbf{C}_{4,2}^T\right)\mathbf{v}_{(t)}^H.\end{aligned} \tag{33}$$

Next, the constraint is recast as

$$\begin{aligned}&\mathbf{v}^H\left(\mathbf{H}_{\mathrm{R},k}\odot\mathbf{C}_{4,1}^T\right)\mathbf{v}-\mathbf{v}_{(t)}^H\left(\mathbf{H}_{\mathrm{R},k}\odot\mathbf{C}_{4,2}^T\right)\mathbf{v}_{(t)}^H\\ &+2\mathrm{Re}\left\{\left[\mathbf{v}_{(t)}^H\left(\mathbf{H}_{\mathrm{R},k}\odot\mathbf{C}_{4,2}^T\right)+\mathbf{c}_3^T\right]\mathbf{v}\right\}+c_k\leq 0,\forall k.\end{aligned} \tag{34}$$

3) Active RIS Transmission Power Constraint: The constraint (18h) is reformulated as

$$\mathrm{Tr}\left(\mathbf{\Psi}^H\mathbf{\Psi}\mathbf{C}_5\right)+\sigma_{\mathrm{R}}^2\mathrm{Tr}\left(\mathbf{\Psi}^H\mathbf{\Psi}\right)\leq P_1, \tag{35}$$

where $\mathbf{C}_5\triangleq\mathbf{H}\mathbf{R}\mathbf{H}^H$. We leverage *Lemma 3* again and obtain the expression in terms of $\mathbf{v}$:

$$\mathbf{v}^H\left[\left(\mathbf{I}_N\odot\mathbf{C}_5^T\right)+\sigma_{\mathrm{R}}^2\mathbf{I}_N\right]\mathbf{v}\leq P_1. \tag{36}$$

4) Problem Reformulation: According to the aforementioned derivations, the active RIS reflecting coefficients matrix optimization problem is recast as

$$\max_{\mathbf{v}}\quad \gamma_{r,lb} \tag{37a}$$

$$\text{s.t.}\quad (34),(36), \tag{37b}$$

$$\mathbf{v}=\left[\psi_1,\cdots,\psi_N\right]^T, \tag{37c}$$

$$\alpha_n\leq\alpha_{\max},1\leq n\leq N. \tag{37d}$$

Problem (37) is a convex quadratically constrained quadratic programming (QCQP) problem and can be solved by using CVX.

C. Optimize $\mathbf{t}$

In this subsection, we select the ports of FAS $\mathbf{t}$ when $\mathbf{R}$ and $\mathbf{\Psi}$ are fixed.

To begin with, we notice that the key challenge to optimize $\mathbf{t}$ arises from the complexity of channel matrix $\mathbf{H}(\mathbf{t})$ and channel vectors $\mathbf{h}_k^H(\mathbf{t})$. Moreover, the coupling of $\{\mathbf{t}_m\}_{m=1}^M$ introduces additional non-convexity. Thus, we transform the problem of port selection into the binary integer programming by reformulating the channel function.

1) Channel Reformulation: We first define an effective channel matrix in the Tx-RIS link, which is expressed as

$$\hat{\mathbf{H}}(\mathbf{p})=\mathbf{F}(\mathbf{r})^H\mathbf{\Sigma}\mathbf{G}(\mathbf{p})\in\mathbb{C}^{N\times Q}, \tag{38}$$

where $\mathbf{p}=\{\mathbf{p}_q\}_{q=1}^Q$ and $\hat{\mathbf{H}}(\mathbf{p})$ collects the channel vectors between a available port of fluid antenna $\mathbf{p}_q$ and the active RIS. Then, the actual channel matrix of the Tx-RIS link is expressed as

$$\mathbf{H}(\mathbf{t})=\hat{\mathbf{H}}(\mathbf{p})\mathbf{B}\in\mathbb{C}^{N\times M}, \tag{39}$$

where $\mathbf{B}=[\mathbf{b}_1,\cdots,\mathbf{b}_M]\in\mathbb{R}^{Q\times M}$ and $\mathbf{b}_m=[b_{1,m},\cdots,b_{Q,m}]^T$ is the port selection vector of the m-th activated port. Besides, $b_{q,m}\in\{0,1\}$ is the element of $\mathbf{b}_m$ and $\sum_{q=1}^Q b_{q,m}=1$. It should be noticed that $b_{q,m}=1$ if and only if the q-th available port is selected as the m-th activated port, i.e. $\mathbf{t}_m=\mathbf{p}_q$. Similar to (38) and (39), the channel vector of the Tx-User link is expressed

$$\begin{aligned}&\frac{\left(\mathbf{h}_k^H+\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{H}\right)\mathbf{R}_k\left(\mathbf{h}_k+\mathbf{H}^H\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}\right)}{\left(\mathbf{h}_k^H+\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{H}\right)\left(\mathbf{R}-\mathbf{R}_k\right)\left(\mathbf{h}_k+\mathbf{H}^H\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}\right)+\sigma_{\mathrm{R}}^2\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}+\sigma_k^2}\geq\gamma_{th}\\ &\Rightarrow\quad f\triangleq\left(\mathbf{h}_k^H+\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{H}\right)\left[\mathbf{R}-\left(1+\gamma_{th}^{-1}\right)\mathbf{R}_k\right]\left(\mathbf{h}_k+\mathbf{H}^H\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}\right)+\sigma_{\mathrm{R}}^2\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}+\sigma_k^2\leq 0.\end{aligned} \tag{30}$$

$$\begin{aligned}f&=2\mathrm{Re}\left\{\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{H}\left[\mathbf{R}-\left(1+\gamma_{th}^{-1}\right)\mathbf{R}_k\right]\mathbf{h}_k\right\}+\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{H}\left[\mathbf{R}-\left(1+\gamma_{th}^{-1}\right)\mathbf{R}_k\right]\mathbf{H}^H\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}\\ &\quad+\sigma_{\mathrm{R}}^2\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}+\mathbf{h}_k^H\left[\mathbf{R}-\left(1+\gamma_{th}^{-1}\right)\mathbf{R}_k\right]\mathbf{h}_k+\sigma_k^2\\ &=2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{\Psi}\mathbf{H}\left[\mathbf{R}-\left(1+\gamma_{th}^{-1}\right)\mathbf{R}_k\right]\mathbf{h}_k\mathbf{h}_{\mathrm{R},k}^H\right)\right\}+\mathrm{Tr}\left(\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\left[\mathbf{H}\left(\mathbf{R}-\mathbf{R}_k\right)\mathbf{H}^H+\sigma_{\mathrm{R}}^2\mathbf{I}_N\right]\right)\\ &\quad+\mathrm{Tr}\left(\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k}\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\left(-\gamma_{\mathrm{th}}^{-1}\mathbf{H}\mathbf{R}_k\mathbf{H}^H\right)\right)+\mathbf{h}_k^H\left[\mathbf{R}-\left(1+\gamma_{th}^{-1}\right)\mathbf{R}_k\right]\mathbf{h}_k+\sigma_k^2\\ &\triangleq 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{\Psi}\mathbf{C}_3\right)\right\}+\mathrm{Tr}\left(\mathbf{\Psi}^H\mathbf{H}_{\mathrm{R},k}\mathbf{\Psi}\left(\mathbf{C}_{4,1}+\mathbf{C}_{4,2}\right)\right)+c_k\leq 0.\end{aligned} \tag{31}$$

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as

$$\mathbf{h}_k^H(\mathbf{t}) = \hat{\mathbf{h}}_k^H(\mathbf{p})\mathbf{B} \in \mathbb{C}^{1\times M}. \tag{40}$$

Besides, the equivalent channel vector between the Tx and the k-th user is rewritten as

$$\begin{aligned}\mathbf{h}_{e,k}^H(\mathbf{t}) &= \mathbf{h}_k^H(\mathbf{t}) + \mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{H}(\mathbf{t})\\ &= \left(\hat{\mathbf{h}}_k^H(\mathbf{p}) + \mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\hat{\mathbf{H}}(\mathbf{p})\right)\mathbf{B}\\ &\triangleq \hat{\mathbf{h}}_{e,k}^H(\mathbf{p})\mathbf{B}.\end{aligned} \tag{41}$$

2) Radar SNR: By utilizing *Lemma 2* without taking the first-order derivative w.r.t. $\mathbf{J}$, we can obtain the approximation of radar SNR at the t-th iteration as

$$\begin{aligned}\gamma_r &= \mathrm{Tr}(\mathbf{A}\mathbf{\Psi}\hat{\mathbf{H}}\mathbf{B}\mathbf{R}\mathbf{B}^H\hat{\mathbf{H}}^H\mathbf{\Psi}^H\mathbf{A}^H\mathbf{J}^{-1})\\ &\geq 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{A}\mathbf{\Psi}\hat{\mathbf{H}}\mathbf{B}\mathbf{R}\mathbf{B}_{(t)}^H\hat{\mathbf{H}}^H\mathbf{\Psi}^H\mathbf{A}^H\mathbf{J}^{-1}\right)\right\}\\ &\quad - \mathrm{Tr}\left(\mathbf{A}\mathbf{\Psi}\hat{\mathbf{H}}\mathbf{B}_{(t)}\mathbf{R}\mathbf{B}_{(t)}^H\hat{\mathbf{H}}^H\mathbf{\Psi}^H\mathbf{A}^H\mathbf{J}^{-1}\right)\\ &\triangleq 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{B}\mathbf{E}_1\right)\right\} - \mathrm{const}_1,\end{aligned} \tag{42}$$

where $\mathbf{E}_1 \triangleq \mathbf{R}\mathbf{B}_{(t)}^H\hat{\mathbf{H}}^H\mathbf{\Psi}^H\mathbf{A}^H\mathbf{J}^{-1}\mathbf{A}\mathbf{\Psi}\hat{\mathbf{H}}$ and the second term of (42) is a constant that does not need to be optimized.

3) Communication SINR Constraint: The communication SINR constraint of the k-th user is transformed into

$$\begin{aligned}\gamma_k &= \frac{\hat{\mathbf{h}}_{e,k}^H\mathbf{B}\mathbf{R}_k\mathbf{B}^H\hat{\mathbf{h}}_{e,k}}{\hat{\mathbf{h}}_{e,k}^H\mathbf{B}(\mathbf{R}-\mathbf{R}_k)\mathbf{B}^H\hat{\mathbf{h}}_{e,k} + \sigma_{\mathrm{R}}^2\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k} + \sigma_k^2}\\ &= \frac{\mathrm{Tr}\left(\mathbf{B}\mathbf{R}_k\mathbf{B}^H\hat{\mathbf{H}}_{e,k}\right)}{\mathrm{Tr}\left(\mathbf{B}(\mathbf{R}-\mathbf{R}_k)\mathbf{B}^H\hat{\mathbf{H}}_{e,k}\right) + \sigma_1^2} \geq \gamma_{th}, \forall k,\end{aligned} \tag{43}$$

where $\hat{\mathbf{H}}_{e,k} \triangleq \hat{\mathbf{h}}_{e,k}\hat{\mathbf{h}}_{e,k}^H$ and $\sigma_1^2 \triangleq \sigma_{\mathrm{R}}^2\mathbf{h}_{\mathrm{R},k}^H\mathbf{\Psi}\mathbf{\Psi}^H\mathbf{h}_{\mathrm{R},k} + \sigma_k^2$. Next, the constraint is rewritten as

$$\mathrm{Tr}\left(\mathbf{B}\bar{\mathbf{R}}_k\mathbf{B}^H\hat{\mathbf{H}}_{e,k}\right) \geq \sigma_1^2, \forall k, \tag{44}$$

where $\bar{\mathbf{R}}_k \triangleq (1+\gamma_{th}^{-1})\mathbf{R}_k - \mathbf{R}$. In order to transform (44) into a convex form, we introduce the follow lemma to tackle it.

Lemma 4: The trace of the product of four matrices can be expressed as

$$\mathrm{Tr}\left(\mathbf{ABCD}\right) = \left(\mathrm{vec}\left(\mathbf{D}^T\right)\right)^T\left(\mathbf{C}^T\otimes\mathbf{A}\right)\mathrm{vec}\left(\mathbf{B}\right). \tag{45}$$

Proof: Please refer to [69].

Thus, (44) can be recast as

$$\left(\mathrm{vec}\left(\mathbf{B}\right)\right)^H\left(\bar{\mathbf{R}}_k^T\otimes\hat{\mathbf{H}}_{e,k}\right)\mathrm{vec}\left(\mathbf{B}\right) \geq \sigma_1^2, \forall k. \tag{46}$$

Note that (46) is still a non-convex constraint. Hence, we introduce the follow lemma to tackle it.

Lemma 5: Constraint (46) is equivalent to the following constraint

$$\begin{aligned}&\left(\mathrm{vec}\left(\mathbf{B}\right)\right)^H\left(-\bar{\mathbf{R}}_k^T\otimes\hat{\mathbf{H}}_{e,k} + \eta\mathbf{I}_{QM}\right)\mathrm{vec}\left(\mathbf{B}\right)\\ &\quad\leq -\sigma_1^2 + \eta M, \forall k,\end{aligned} \tag{47}$$

where $\eta \in \mathbb{R}_+$ is a arbitrary positive real number.

Proof: Please refer to Appendix B.

It should be noticed that when $\eta \geq \lambda_{\max}(\bar{\mathbf{R}}_k^T\otimes\hat{\mathbf{H}}_{e,k})$, constraint (47) is convex.

4) Active RIS Transmission Power Constraint: Constraint (18h) can be rewritten as

$$\mathrm{Tr}\left(\mathbf{B}\mathbf{R}\mathbf{B}^H\mathbf{E}_2\right) \leq \mathrm{const}_2, \tag{48}$$

where $\mathbf{E}_2 \triangleq \hat{\mathbf{H}}^H\mathbf{\Psi}^H\mathbf{\Psi}\hat{\mathbf{H}}$ and $\mathrm{const}_2 \triangleq P_1 - \boldsymbol{\sigma}_{\mathrm{R}}^2\mathrm{Tr}(\mathbf{\Psi}\mathbf{\Psi}^H)$. Similar to (44), (48) can be transformed into a convex constraint as follows

$$\left(\mathrm{vec}\left(\mathbf{B}\right)\right)^H\left(\mathbf{R}^T\otimes\mathbf{E}_2\right)\mathrm{vec}\left(\mathbf{B}\right) \leq \mathrm{const}_2. \tag{49}$$

5) Position and Distance Constraints of FAS: Based on the previous derivations, the position constraint for FAS is given by

$$\begin{cases}\sum_{q=1}^{Q} b_{q,m} = 1, 1 \leq m \leq M,\\ b_{q,m} \in \{0,1\}, 1 \leq q \leq Q, 1 \leq m \leq M.\end{cases} \tag{50}$$

To denote the distance between any pair of fluid antennas, we define a distance matrix $\mathbf{D} \in \mathbb{R}^{Q\times Q}$, with $[\mathbf{D}]_{q,q'}$ represents the distance between $\mathbf{p}_q$ and $\mathbf{p}_{q'}$. Thus, the distance constraint can be reformulated as

$$\mathbf{b}_m^T\mathbf{D}\mathbf{b}_{m'} \geq D_{\min}, 1 \leq m, m' \leq M, m \neq m'. \tag{51}$$

To simplify the derivation and computation, we transform (51) into the following form

$$\left(\mathrm{vec}\left(\mathbf{B}\right)\right)^H\left(\frac{\mathbf{D}_{m,m'} + \mathbf{D}_{m,m'}^T}{2}\right)\mathrm{vec}\left(\mathbf{B}\right) \geq D_{\min}, \tag{52}$$

where $\mathbf{D}_{m,m'} = \mathbf{V}_{m,m'}\otimes\mathbf{D}$ and $\mathbf{V}_{m,m'} \in \mathbb{R}^{M\times M}$ with $[\mathbf{V}_{m,m'}]_{m,m'} = 1$ and other elements are 0. By utilizing *Lemma 5*, (52) can be recast as

$$\begin{aligned}&\left(\mathrm{vec}\left(\mathbf{B}\right)\right)^H\left(-\frac{\mathbf{D}_{m,m'} + \mathbf{D}_{m,m'}^T}{2} + \eta\mathbf{I}_{QM}\right)\mathrm{vec}\left(\mathbf{B}\right)\\ &\quad\leq -D_{\min} + \eta M.\end{aligned} \tag{53}$$

6) Problem Reformulation: Following the previous deduction, the port selection problem of FAS is reformulated as

$$\max_{\mathbf{B}} \quad 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{B}\mathbf{E}_1\right)\right\} \tag{54a}$$

$$\text{s.t.} \quad (47), (49), (53), \tag{54b}$$

$$\sum_{q=1}^{Q} b_{q,m} = 1, 1 \leq m \leq M, \tag{54c}$$

$$b_{q,m} \in \{0,1\}, 1 \leq q, q' \leq Q, 1 \leq m, m' \leq M. \tag{54d}$$

Note that (54d) is a binary constraint, which renders Problem (54) non-convex. To handle the difficulty, we transform (54d) into the following inequality constraints

$$\sum_{q=1}^{Q}\sum_{m=1}^{M}\left(b_{q,m} - b_{q,m}^2\right) \leq 0, \tag{55}$$

$$0 \leq b_{q,m} \leq 1, \quad 1 \leq q, q' \leq Q, 1 \leq m, m' \leq M. \tag{56}$$

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Thus, Problem (54) is rewritten as

$$\max_{\mathbf{B}} \quad 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{B}\mathbf{E}_1\right)\right\} \tag{57a}$$

$$\text{s.t.} \quad (47), (49), (53), (54c), (55), (56). \tag{57b}$$

By applying the penalty-based method, we regard (55) as a penalty term and incorporate it into the objective function. Then, we rewrite Problem (57) as follows

$$\max_{\mathbf{B}} \quad 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{B}\mathbf{E}_1\right)\right\} + \mu g\left(\mathbf{B}\right) \tag{58a}$$

$$\text{s.t.} \quad (47), (49), (53), (54c), (56), \tag{58b}$$

where $g(\mathbf{B}) \triangleq \sum_{q=1}^{Q}\sum_{m=1}^{M}(b_{q,m}^2 - b_{q,m})$ and $\mu > 0$ is the penalty factor. It is obvious that when μ is sufficiently large, i.e. $\mu \to +\infty$, the value of $g(\mathbf{B})$ approaches 0. Note that the objective function of (58) is non-convex, thus, we construct a lower bound of $g(\mathbf{B})$ by adopting the SCA method

$$\hat{g}\left(\mathbf{B}\right) = \sum_{q=1}^{Q}\sum_{m=1}^{M}\left(2b_{q,m,(t)}b_{q,m} - b_{q,m,(t)}^2 - b_{q,m}\right), \tag{59}$$

where $b_{q,m,(t)}$ denotes the value for $b_{q,m}$ in the t-th iteration. Finally, Problem (54) is reformulated as

$$\max_{\mathbf{B}} \quad 2\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{B}\mathbf{E}_1\right)\right\} + \mu\hat{g}\left(\mathbf{B}\right) \tag{60a}$$

$$\text{s.t.} \quad (47), (49), (53), (54c), (56). \tag{60b}$$

Problem (60) is a convex problem and can be solved by using CVX.

D. Convergence and Complexity Analysis

According to above computation and derivations, the overall algorithm is concluded in Algorithm 1. The proposed algorithm is based on the AO and MM algorithm. Thus, the solution to Problem (18) is continuously increasing in each iteration of Algorithm 1. Moreover, the solution to Problem (18) has an upper bound as a consequence of the transmission power constraint. Thus, the convergence of Algorithm 1 is guaranteed.

Next, we analyze the complexity of the proposed algorithm. Firstly, the approximate complexity of Problem (22) is $\mathcal{O}(K^{3.5}M^{6.5}\ln(1/\epsilon))$ [70], where $\epsilon > 0$ denotes the solution accuracy. Then, the approximate complexity of solving Problem (37) is $\mathcal{O}(t_1^{\max}o_a\ln(1/\epsilon))$, where $o_a = (K+N)^{0.5}KN^3$. In addition, the approximate complexity of solving Problem (60) is $\mathcal{O}(t_2^{\max}o_b\ln(1/\epsilon))$, where $o_b = (K+QM)^{0.5}KQ^3M^3$. Thus, the complexity of proposed overall algorithm is $\mathcal{O}(t_{\mathrm{AO}}^{\max}(K^{3.5}M^{6.5} + t_1^{\max}o_a + t_2^{\max}o_b)\ln(1/\epsilon))$.

IV. Numerical Results

In this section, numerical results are presented to evaluate the performance of proposed FAS and active RIS-aided ISAC system.

A. Parameter Settings

Unless otherwise specified, the simulation parameters are defined as follows. As shown in Fig. 2, we assume that the Tx, the RIS, the target and the Rx are located in (0, 0), (20 m, 10 m), (30 m, 0) and (40 m, 0), respectively. In addition, the users are randomly and uniformly distributed within a circle centered at (10 m, -20 m) with the radius of 10 meters. For the channel model, we consider the geometric channel and $L_R^t = L_R^r = L_U^{t,k} = L_U^{r,k} = L_p, \forall k$ [32]. Then, the Rician channel model is adopted for the Tx-RIS and RIS-User links. Furthermore, the PRMs of Rician channels are expressed as

$$[\mathbf{\Sigma}]_{1,1} \sim \mathcal{CN}\left(0, \frac{\beta G_0 (d/d_0)^{-\alpha}}{\beta+1}\right),$$

Algorithm 1: Proposed AO-based Algorithm for Solving Problem (18).

Input: The maximum iteration time $t_1^{\max}$, $t_2^{\max}$ and $t_{\mathrm{AO}}^{\max}$ for Problem (37), Problem (60) and Problem (18). The solution accuracy ϵ_1, ϵ_2 and ϵ_{AO} for Problem (37), Problem (60) and Problem (18).
Set $t_{\mathrm{AO}} = 0$, initialize $\mathbf{\Psi}$ and $\mathbf{B}$.
Repeat
  Update $\mathbf{R}$ and $\{\mathbf{R}_k\}$ by solving Problem (21).
  Set $t_1 = 0$.
  Repeat
    Update $\mathbf{\Psi}$ by solving Problem (37).
    $t_1 = t_1 + 1$.
  Until $t_1 = t_1^{\max}$ or the increase of the objective function is below ϵ_1.
  Set $t_2 = 0$.
  Repeat
    Update $\mathbf{B}$ by solving Problem (60).
    $t_2 = t_2 + 1$.
  Until $t_2 = t_2^{\max}$ or the increase of the objective function is below ϵ_2.
  $t_{\mathrm{AO}} = t_{\mathrm{AO}} + 1$.
Until $t_{\mathrm{AO}} = t_{\mathrm{AO}}^{\max}$ or the increase of the objective function is below ϵ_{AO}.

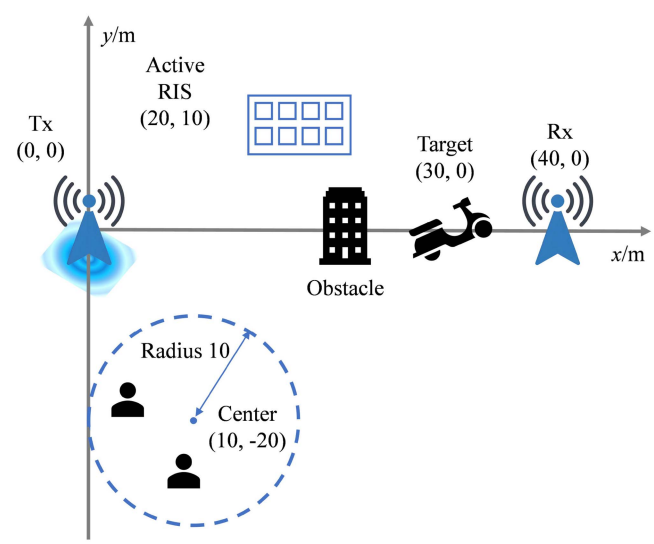

Fig. 2. The simulation system setup.

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TABLE II
THE SIMULATION PARAMETERS

Parameter	Description	Value
f_c	Frequency of carrier	6 GHz
P_0	Maximum transmit power at Tx	40 dBm
Q	The number of available ports	64
M	The number of activated ports	4
A	Size of movable region	2λ
$D_{\min}$	Minimum antenna distance	0.5λ
L_p	The number of channel paths	4
β	Rician factor	0.5
G_0	The path loss at the reference distance	-40 dB
d_0	The reference distance	1 m
α_{TU}	The path loss exponent for Tx-User link	3.5
α_{TR}	The path loss exponent for Tx-RIS link	2.5
α_{RU}	The path loss exponent for RIS-User link	2.5
N	The number of RIS elements	16
$\alpha_{\max}$	Maximum amplification factor	100
P_1	Maximum transmit power at RIS	10 dBm
σ_{R}^2	AWGN at RIS	-80 dBm
K	The number of users	2
γ_{th}	The SINR threshold	10 dB
σ_k^2	AWGN at users	-80 dBm
L	The number of antennas at Rx	2
σ_r^2	AWGN at Rx	-80 dBm

$$[\mathbf{\Sigma}]_{\delta,\delta} \sim \mathcal{CN}\left(0, \frac{G_0(d/d_0)^{-\alpha}}{(\beta+1)(L_p-1)}\right), \delta = 2, \cdots, L_p,$$

where β is the Rician factor and $G_0(d/d_0)^{-\alpha}$ is the path loss. Among which, G_0 represents the path loss at the reference distance d_0, d is the propagation distance, and α is the exponent of path loss. Since the communication users are usually surrounded by obstructions, we adopt the Rayleigh channel to model the Tx-User link and the standard deviation of shadow fading is set to 3dB. Besides, the elevation and azimuth angles for all channel paths are independent and each of them follows the uniform distribution across the interval $[0, \pi]$. All simulation parameters are listed in Table II. Besides, each numerical result is obtained by averaging over 200 independent channel realizations.

B. Convergence Behaviors

In this subsection, we assess the convergence performance of the proposed algorithm in some different conditions. As illustrated in Fig. 3, our algorithm can achieve convergence within 20 iterations. Besides, the increasing of objective function is negligible after 10 iterations, which validate the effective convergence behavior of the algorithm. Furthermore, it is observed that the FAS-aided system exhibits a performance enhancement of approximately 1 dB compared to the FPA-aided counterpart, which demonstrates the superiority of enabling FAS in ISAC systems. In addition, the number of RIS elements has negligible impact on convergence behavior.

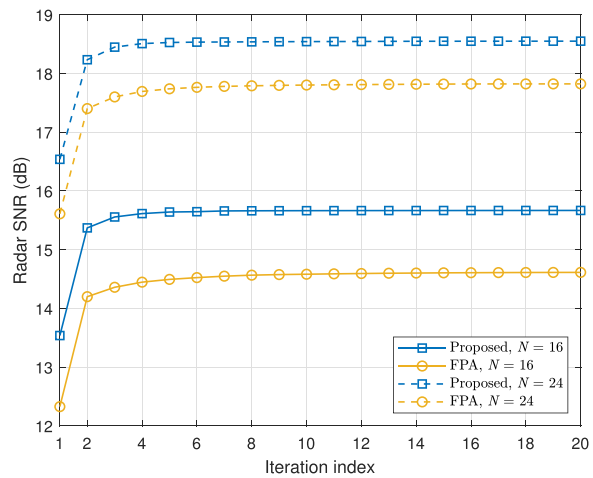

Fig. 3. Convergence Behaviors.

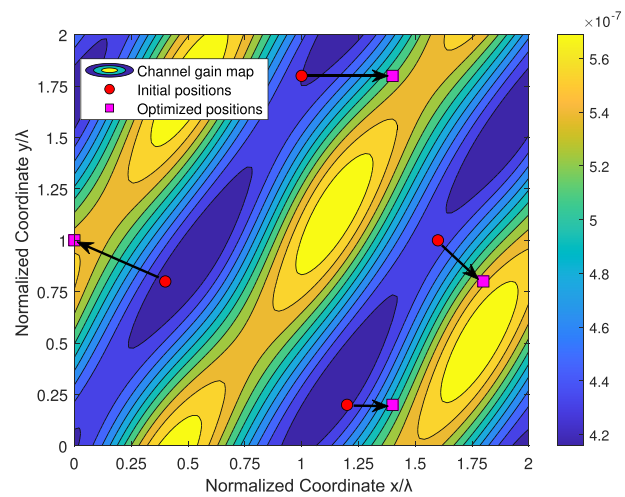

Fig. 4. The channel gain map of Tx-RIS link versus the movement of fluid antennas.

C. The Relationship Between Channel Gain and FAS

Fig. 4 illustrates a realization of the channel gain for Tx-RIS link versus the positions of FAS. For a clearer visualization of the channel gain map, we use a finer grid with $Q = 100$ in this figure, while all other simulations adhere to $Q = 64$ as listed in Table II. Besides, the channel gain is defined as $\|\mathbf{F}(\mathbf{r})^H \mathbf{\Sigma} \mathbf{g}(\mathbf{t}_0)\|_2^2$, where $\mathbf{t}_0$ represents the coordinates of an antenna. We can observe that the positions of FAS move from low-gain to high-gain regions, revealing how FAS fully exploits the spatial variation of channel characteristics. Moreover, it is noticed that the fluid antennas do not move to the region with maximum channel gain, which can be attribute to three main reasons. Firstly, the discrete positions of FAS restrict the freedom movement of fluid antennas. Secondly, the ISAC system needs to realize the trade-off between the communication and sensing performance.

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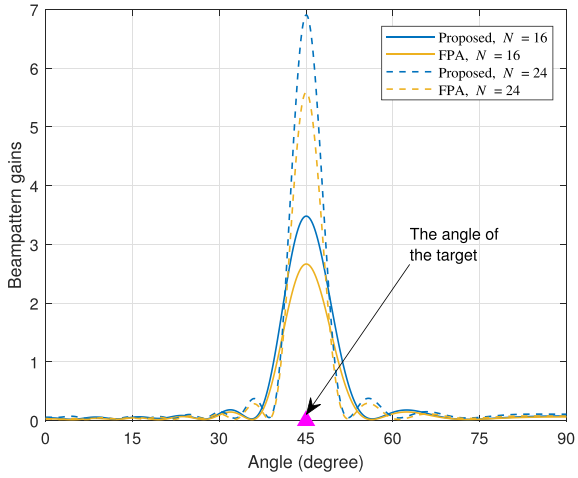

Fig. 5. The beampattern gains in different scenarios.

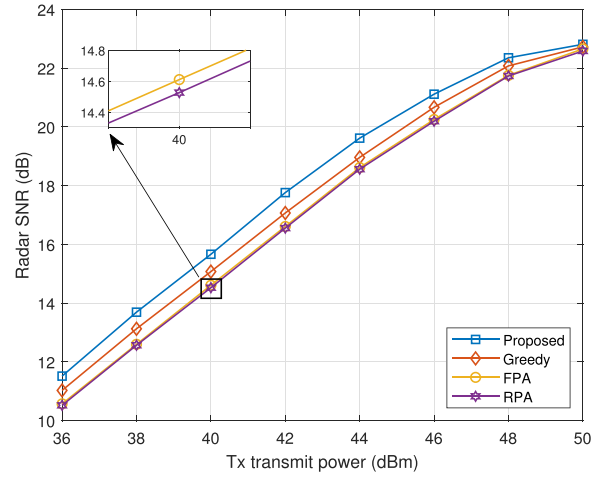

Fig. 6. Radar SNR versus the transmit power at Tx P_0.

Lastly, there exist LoS and reflecting communication channels simultaneously. The regions with high-gain in the Tx-RIS link are likely to have lower gain in the Tx-User link.

D. Beampattern of the Proposed System

Fig. 5 illustrates the beampattern gains at the active RIS for different conditions. The beampattern gains at the active RIS can be expressed as

$$\begin{aligned}\mathcal{P}(\theta) &= \mathbb{E}\left[\left|\mathbf{a}_1^H(\theta)\,\mathbf{\Psi}\mathbf{H}\mathbf{x}\right|^2\right] \\ &= \mathbf{a}_1^H(\theta)\,\mathbf{\Psi}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}^H\mathbf{a}_1(\theta), \end{aligned} \tag{61}$$

where θ represents the AoD w.r.t. the active RIS. It is observed that the FAS-enabled system exhibits 30% enhancement in beampattern gains at the angle of the target compared to the FPA-enabled system. Furthermore, increasing the number of RIS elements not only amplifies the beampattern gains, but also enhances the spatial resolution by narrowing the major lobe, which demonstrates the benefit of active RIS in enhancing the spatial directionality and sensing performance.

E. Performance Comparison With Baseline Schemes

In order to fully assess the performance and demonstrate the advantages of enabling FAS in an active RIS-aided system, we compare the radar SNR of our FAS-aided system with the following three benchmarks:

1) *Benchmark 1. Greedy Algorithm (Greedy):* In this benchmark, we select the ports of antennas sequentially, until all the M ports are selected.
2) *Benchmark 2. FPA:* In this benchmark, the Tx deploys an FPA-based ULA with M antennas spaced at the interval of 0.5λ.
3) *Benchmark 3. Random position antenna (RPA):* In this benchmark, the positions of M antennas are randomly generated, subject to the minimum distance constraint.

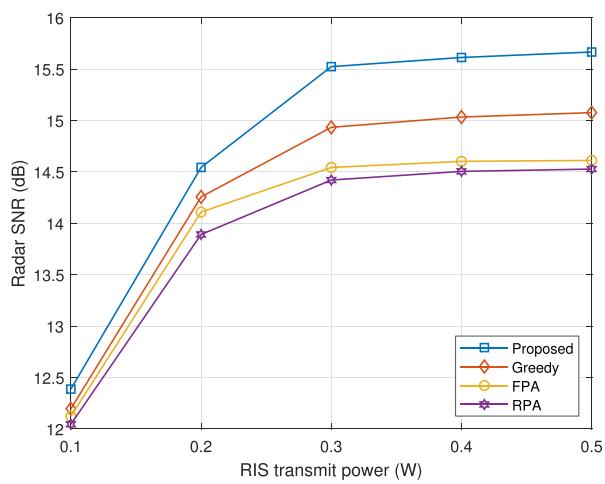

Fig. 7. Radar SNR versus the transmit power at the RIS P_1.

Fig. 6 presents the relationship between radar SNR and the transmit power at the Tx. We can observe that the proposed algorithm exhibits approximately 0.5 dB performance gain compared to the greedy algorithm, which demonstrates the superiority of our algorithm. Moreover, the proposed algorithm can achieve the same sensing performance with lower power, which indicates the benefit of FAS in reducing transmit power. Furthermore, it is noticed that the growth rate of radar SNR begins to diminish when $P_0 = 46$ dBm. That is reasonable since the signal power at the active RIS increases with the growth of P_0. Thus, the transmit power at RIS is likely to reach its maximum value P_1, which constrains the further amplification of reflected signal.

The relationship between radar SNR and the transmit power at the RIS P_1 is presented in Fig. 7. As depicted in Fig. 7, the growth rate of radar SNR begins to diminish when $P_1 = 0.3$ W, which can be attribute to the limitation of maximum amplification factor $\alpha_{\max}$. Besides, it is observed that the FAS-aided system can achieve the same performance with less transmit power at

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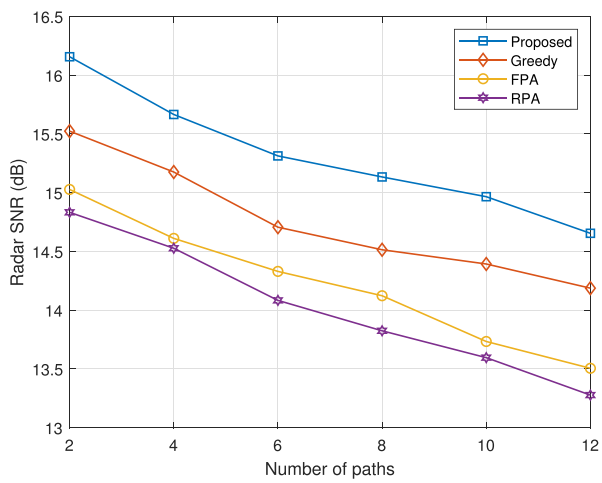

Fig. 8. Radar SNR versus the number of channel paths L_p.

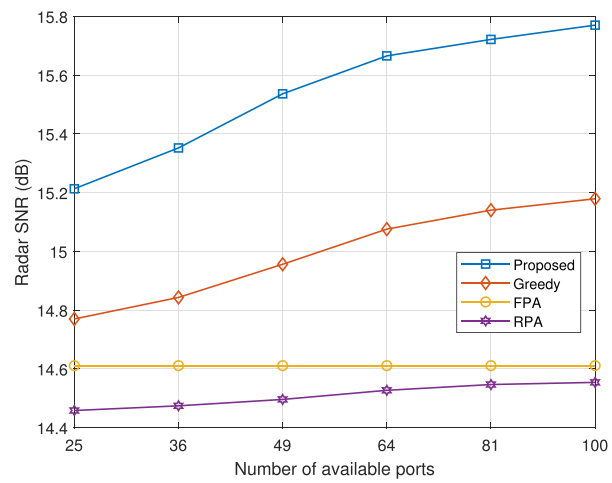

Fig. 10. Radar SNR versus the number of available ports Q.

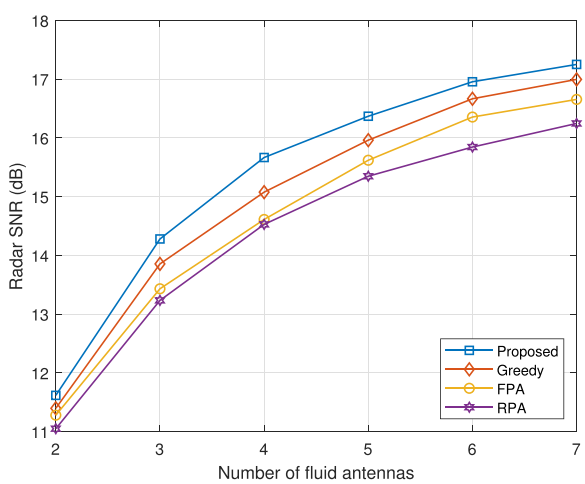

Fig. 9. Radar SNR versus the number of fluid antennas M.

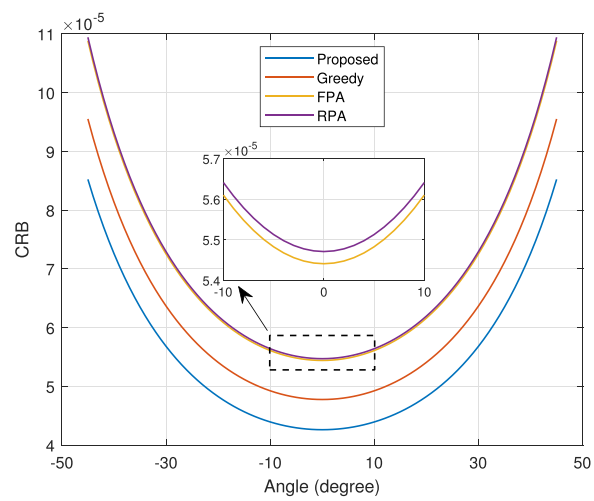

Fig. 11. CRB for estimating θ_2.

the RIS, which reveals the superiority of FAS in reducing the power at the active RIS.

Fig. 8 depicts the relationship between the radar SNR and the number of channel paths. As illustrated in Fig. 8, the increase of channel paths L_p results in the deterioration of sensing performance. This phenomenon contrasts with prior researches [49], [50], [51] where channel capacity improves with channel paths. The decline of sensing performance can be attributed to the scattering of the electromagnetic wave [59], [71].

Fig. 9 depicts the radar SNR of the proposed and benchmark schemes versus the number of fluid antennas. It is illustrated that the radar SNR exhibits an enhancement as the number of antennas M increases. This phenomenon can be ascribed to the extra DoFs offered by additional antennas. Besides, the FAS-aided system requires less antennas to achieve the same performance, which demonstrates the advantage of FAS in reducing the number of antennas.

Fig. 10 presents the impact of the number of available ports Q on the sensing performance. As expected, the radar SNR improves with the number of available ports. This can be attributed to the increment of spatial DoFs. Besides, the growth rate of sensing performance decreases with the rise of Q, which indicates that a moderate number of available ports can achieve adequate performance.

To measure the accuracy of estimating target's parameters, we calculate the CRB for estimating θ_2 in Appendix C and illustrate the variation of CRB in Fig. 11. The number of transmit symbols is set as $T = 128$ to ensure it is large enough. It is observed that the all schemes reach the minimum value at the expected angle and FAS can reduce the CRB for estimating the angle of the target.

F. Performance Comparison With Passive RIS

In this subsection, we demonstrate the superiority of the active RIS-aided system by comparing the proposed scheme with the passive RIS scheme. For the fairness of comparison, we consider the same total power allocation. The power budgets of active and

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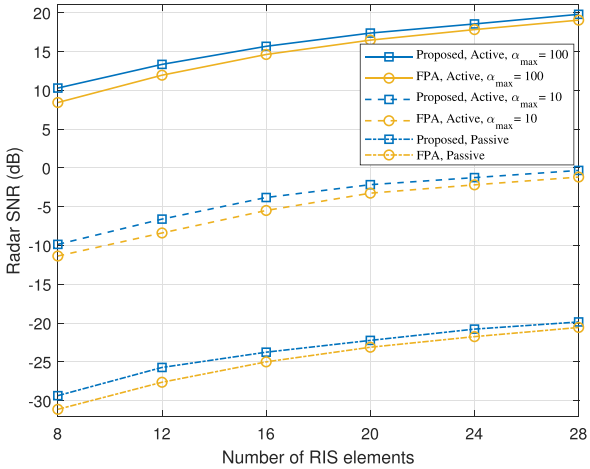

Fig. 12. Radar SNR versus the number of RIS elements N.

passive RIS are as follows

$$P^{\text{act}} = P_0^{\text{act}} + P_1 + N\left(P_{\text{SW}} + P_{\text{DC}}\right),$$
$$P^{\text{pas}} = P_0^{\text{pas}} + NP_{\text{SW}}, \tag{62}$$

where P^{act} and P^{pas} represent the total power of active and passive RIS schemes, respectively. Besides, P_0^{act} and P_0^{pas} denote the transmit power at the Tx for active and passive RIS schemes, respectively. Next, the parameter $P_{\text{SW}} = -5$ dBm is the power utilized for adjusting phase in each RIS element and $P_{\text{DW}} = -10$ dBm represents the biasing power of each active RIS element [19]. For the active RIS scheme, the transmit power of active RIS is set as $P_1 = 1$ W and the total power is set as $P^{\text{act}} = 11$ W. In addition, the remaining power is allocated to P_0^{act}. For the passive RIS scheme, we set the total power as $P^{\text{pas}} = P^{\text{act}} = 11$ W and the remaining power is also allocated to P_0^{pas}.

According to aforementioned parameter settings, Fig. 12 illustrates the relationship between sensing performance and the number of RIS elements. As illustrated in Fig. 12, the performance of the FAS-aided system with fewer RIS elements can match that of FPA-aided system. Moreover, the active RIS schemes significantly outperform their passive RIS counterparts with more RIS elements, which demonstrates that increasing the number of elements cannot improve the performance significantly for passive RIS. This phenomenon also reveals the advantage of active RIS in overcoming path loss.

V. Conclusion

This paper investigated a FAS-enabled and active RIS-aided ISAC system, where the LoS link between Tx and the radar target is blocked. To improve the overall performance of the ISAC system, we formulated an optimization problem to maximize the radar SNR by jointly optimizing the beamforming matrix at the Tx, the reflecting coefficients of active RIS and the port selection of FAS. Given that the problem is highly non-convex and the variables are tightly coupled, we proposed an AO-based algorithm to divide the main problem into three subproblems. Specifically, we first leveraged the SDR method to optimize the beamforming matrix, then, the MM method was adopted to optimize the reflecting coefficients of active RIS. For the port selection of FAS, we first transformed the subproblem into binary programming, then the penalty function and SCA method were applied to tackle it. Our numerical results illustrated the superiority of the FAS and active RIS-aided ISAC system, where the enabling of FAS reduces the resource consumption (e.g. the number of antennas, the number of RIS elements and the transmit power) and the assistance of active RIS overcomes severe path loss effectively. In addition, existing researches mainly focus on sensing single point target. Nevertheless, sensing extended target or multi targets is a more practical scenario and its implementation in FAS and active RIS-aided ISAC systems remains unexplored.

Appendix A
Proof for Lemma 2

It can be proved that $f(\mathbf{L}, \mathbf{L}^*, \mathbf{J}) = \text{Tr}(\mathbf{L}^H\mathbf{J}^{-1}\mathbf{L})$ is jointly convex w.r.t. $\mathbf{L}$ and $\mathbf{J}$. We apply the first-order Taylor expansion of $f(\mathbf{L}, \mathbf{L}^*, \mathbf{J})$ at $\mathbf{L}_{(t)}$ and $\mathbf{J}_{(t)}$ to construct the following inequality

$$\begin{aligned} f\left(\mathbf{L}, \mathbf{L}^*, \mathbf{J}\right) \geq\ & f\left(\mathbf{L}_{(t)}, \mathbf{L}^*_{(t)}, \mathbf{J}_{(t)}\right) \\ & + \text{Tr}\left(\left(\frac{\partial f}{\partial \mathbf{L}}\bigg|_{\mathbf{L}=\mathbf{L}_{(t)}}\right)^T \left(\mathbf{L} - \mathbf{L}_{(t)}\right)\right) \\ & + \text{Tr}\left(\left(\frac{\partial f}{\partial \mathbf{L}^*}\bigg|_{\mathbf{L}^*=\mathbf{L}^*_{(t)}}\right)^T \left(\mathbf{L}^* - \mathbf{L}^*_{(t)}\right)\right) \\ & + \text{Tr}\left(\left(\frac{\partial f}{\partial \mathbf{J}}\bigg|_{\mathbf{J}=\mathbf{J}_{(t)}}\right)^T \left(\mathbf{J} - \mathbf{J}_{(t)}\right)\right). \end{aligned} \tag{63}$$

According to [69], the first-order derivation of $f(\mathbf{L}, \mathbf{L}^*, \mathbf{J})$ can be computed as

$$\begin{aligned} \frac{\partial f}{\partial \mathbf{L}}\bigg|_{\mathbf{L}=\mathbf{L}_{(t)}} &= \mathbf{J}^{-T}\mathbf{L}^*_{(t)}, \\ \frac{\partial f}{\partial \mathbf{L}^*}\bigg|_{\mathbf{L}^*=\mathbf{L}^*_{(t)}} &= \mathbf{J}^{-1}\mathbf{L}_{(t)}, \\ \frac{\partial f}{\partial \mathbf{J}}\bigg|_{\mathbf{J}=\mathbf{J}_{(t)}} &= -\left(\mathbf{J}^{-1}_{(t)}\mathbf{L}\mathbf{L}^H\mathbf{J}^{-1}_{(t)}\right)^T. \end{aligned} \tag{64}$$

Thus, by substituting the above derivations into (63), we can calculate the lower bound of $f(\mathbf{L}, \mathbf{L}^*, \mathbf{J})$ as

$$\begin{aligned} f\left(\mathbf{L}, \mathbf{L}^*, \mathbf{J}\right) &= \text{Tr}\left(\mathbf{L}^H\mathbf{J}^{-1}\mathbf{L}\right) \\ &\geq 2\text{Re}\left\{\text{Tr}\left(\mathbf{L}^H_{(t)}\mathbf{J}^{-1}_{(t)}\mathbf{L}\right)\right\} \\ &\quad - \text{Tr}\left(\mathbf{J}^{-1}_{(t)}\mathbf{L}_{(t)}\mathbf{L}^H_{(t)}\mathbf{J}^{-1}_{(t)}\mathbf{J}\right). \end{aligned} \tag{65}$$

Hence, the proof is completed.

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Appendix B
Proof for Lemma 5

By adding a negative sign on both sides of (46), we can obtain

$$(\mathrm{vec}\,(\mathbf{B}))^H \left(-\bar{\mathbf{R}}_k^T \otimes \hat{\mathbf{H}}_{e,k}\right) \mathrm{vec}\,(\mathbf{B}) \leq -\sigma_1^2, \forall k. \tag{66}$$

Next, according to the position and distance constraints (50), we can obtain the following equation

$$(\mathrm{vec}\,(\mathbf{B}))^H \,\mathrm{vec}\,(\mathbf{B}) = \mathrm{Tr}\left(\mathbf{B}^H\mathbf{B}\right) = M. \tag{67}$$

Finally, by adding $\eta(\mathrm{vec}(\mathbf{B}))^H\mathrm{vec}(\mathbf{B})$ on both sides of (66), the constraint is recast as

$$\begin{aligned}&(\mathrm{vec}\,(\mathbf{B}))^H \left(-\bar{\mathbf{R}}_k^T \otimes \hat{\mathbf{H}}_{e,k} + \eta\mathbf{I}_{QM}\right) \mathrm{vec}\,(\mathbf{B})\\ &\leq -\sigma_1^2 + \eta M, \forall k.\end{aligned} \tag{68}$$

Hence, the proof is completed.

Appendix C
Derivation of CRB

By transmitting T symbols $\mathbf{X} = [\mathbf{x}(1), \cdots, \mathbf{x}(T)]$, the echo signal at the Rx is formulated as

$$\begin{aligned}\mathbf{Y}_r &= \mathbf{A}\mathbf{\Psi}(\mathbf{H}\mathbf{X} + \mathbf{N}_\mathrm{R}) + \mathbf{N}_r\\ &= \mathbf{A}\mathbf{\Psi}\mathbf{H}\mathbf{X} + \mathbf{A}\mathbf{\Psi}\mathbf{N}_\mathrm{R} + \mathbf{N}_r,\end{aligned} \tag{69}$$

where $\mathbf{Y}_r$, $\mathbf{N}_\mathrm{R}$ and $\mathbf{N}_r$ have the similar format with $\mathbf{X}$. Thus, the vectorized signal is given by

$$\tilde{\mathbf{y}}_r = \mathbf{p}_s + \mathbf{p}_\mathrm{R} + \mathbf{p}_r, \tag{70}$$

where $\mathbf{p}_s \triangleq \mathrm{vec}(\mathbf{A}\mathbf{\Psi}\mathbf{H}\mathbf{X})$, $\mathbf{p}_\mathrm{R} \triangleq \mathrm{vec}(\mathbf{A}\mathbf{\Psi}\mathbf{N}_\mathrm{R})$ and $\mathbf{p}_r \triangleq \mathrm{vec}(\mathbf{N}_r)$. Besides, the equivalent noise can be denoted as $\tilde{\mathbf{n}} = \mathbf{p}_\mathrm{R} + \mathbf{p}_r$, which follows $\mathcal{CN}(\mathbf{0}, \mathbf{I}_T \otimes \mathbf{J})$.

The estimation parameters are represented as $\xi \triangleq [\theta_2, \boldsymbol{\beta}_p^T]^T$ with $\boldsymbol{\beta}_p \triangleq [\mathrm{Re}(\beta_p), \mathrm{Im}(\beta_p)]^T$. Thus, the Fisher information matrix can be expressed as

$$\mathbf{M} = \begin{bmatrix}\mathbf{M}_{\theta_2\theta_2} & \mathbf{M}_{\theta_2\boldsymbol{\beta}_p}\\ \mathbf{M}_{\theta_2\boldsymbol{\beta}_p}^T & \mathbf{M}_{\boldsymbol{\beta}_p\boldsymbol{\beta}_p}\end{bmatrix}. \tag{71}$$

Since $\mathbf{p}_\mathrm{R}$ is much less than $\mathbf{p}_s$, $\tilde{\mathbf{n}}$ can be assumed to be independent with θ_2. According to [46], each element of $\mathbf{M}$ can be calculated as

$$\mathbf{M}_{\theta_2\theta_2} = 2Ta_1\left|\beta_p\right|^2 \dot{\mathbf{a}}_2^H\mathbf{J}^{-1}\dot{\mathbf{a}}_2, \tag{72}$$

$$\mathbf{M}_{\theta_2\boldsymbol{\beta}_p} = 2Ta_1\mathrm{Re}\left(\beta_p^*\dot{\mathbf{a}}_2^H\mathbf{J}^{-1}\mathbf{a}_2\left[1, j\right]\right), \tag{73}$$

$$\mathbf{M}_{\boldsymbol{\beta}_p\boldsymbol{\beta}_p} = 2Ta_1\mathbf{a}_2^H\mathbf{J}^{-1}\mathbf{a}_2\mathbf{I}_2. \tag{74}$$

where $a_1 = \mathbf{a}_1^H\mathbf{\Psi}\mathbf{H}\mathbf{R}\mathbf{H}^H\mathbf{\Psi}^H\mathbf{a}_1$ and $\dot{\mathbf{a}}_2 = \frac{\partial\mathbf{a}_2}{\partial\theta_2}$. Besides, the l-th element of $\dot{\mathbf{a}}_2$ is expressed as

$$\left[\dot{\mathbf{a}}_2\right]_l = j\frac{2\pi d_\mathrm{Rx}}{\lambda}(l-1)\cos\theta_2 e^{j\frac{2\pi d_\mathrm{Rx}}{\lambda}(l-1)\sin\theta_2}. \tag{75}$$

Thus, the CRB for estimating θ_2 is expressed as

$$\mathrm{CRB}\,(\theta_2) = \left[\mathbf{M}^{-1}\right]_{1,1}. \tag{76}$$

References

[1] F. Liu et al., "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE J. Sel. Areas Commun.*, vol. 40, pp. 1728–1767, Jun. 2022.

[2] F. Liu, C. Masouros, A. P. Petropulu, H. Griffiths, and L. Hanzo, "Joint radar and communication design: Applications state-of-the-art and the road ahead," *IEEE Trans. Commun.*, vol. 68, no. 6, pp. 3834–3862, Jun. 2020.

[3] Z. Ren, Y. Yu, H. Ren, C. Pan, and J. Wang, "What is the optimal inter-site distance in multi-BS cooperative sensing?," *Sci. China Inf. Sci.*, vol. 67, no. 12, Dec. 2024, Art. no. 229302.

[4] J. Tang et al., "Cooperative ISAC-empowered low-altitude economy," *IEEE Trans. Wireless Commun.*, vol. 24, no. 5, pp. 3837–3853, May 2025.

[5] F. Liu, C. Masouros, A. Li, H. Sun, and L. Hanzo, "MU-MIMO communications with MIMO radar: From co-existence to joint transmission," *IEEE Trans. Wireless Commun.*, vol. 17, no. 4, pp. 2755–2770, Apr. 2018.

[6] M. Labib, V. Marojevic, A. F. Martone, J. H. Reed, and A. I. Zaghloui, "Coexistence between communications and radar systems: A survey," *URSI Radio Sci. Bull.*, vol. 2017, no. 362, pp. 74–82, 2017.

[7] C. Pan et al., "Reconfigurable intelligent surfaces for 6G systems: Principles, applications, and research directions," *IEEE Commun. Mag.*, vol. 59, no. 6, pp. 14–20, Jun. 2021.

[8] M. A. ElMossallamy, H. Zhang, L. Song, K. G. Seddik, Z. Han, and G. Y. Li, "Reconfigurable intelligent surfaces for wireless communications: Principles, challenges, and opportunities," *IEEE Trans. Cogn. Commun. Netw.*, vol. 6, no. 3, pp. 990–1002, Sep. 2020.

[9] C. Pan et al., "An overview of signal processing techniques for RIS/IRS-aided wireless systems," *IEEE J. Sel. Topics Signal Process.*, vol. 16, pp. 883–917, Aug. 2022.

[10] Y. Liu et al., "Reconfigurable intelligent surfaces: Principles and opportunities," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1546–1577, thirdquarter 2021.

[11] Q. Wu and R. Zhang, "Towards smart and reconfigurable environment: Intelligent reflecting surface aided wireless network," *IEEE Commun. Mag.*, vol. 58, no. 1, pp. 106–112, Jan. 2020.

[12] G. Zhou, C. Pan, H. Ren, P. Popovski, and A. L. Swindlehurst, "Channel estimation for RIS-aided multiuser millimeter-wave systems," *IEEE Trans. Signal Process.*, vol. 70, pp. 1478–1492, 2022.

[13] Y. Jin, Z. Yu, R. Weng, B. Wang, H. Ren, and C. Pan, "Reconfigurable intelligent surface-aided dual-function radar and communication system with MU-MIMO communication," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Apr. 2024, pp. 1–6.

[14] G. Zhou, C. Pan, H. Ren, K. Wang, and Z. Peng, "Secure wireless communication in RIS-aided MISO system with hardware impairments," *IEEE Wireless Commun. Lett.*, vol. 10, no. 6, pp. 1309–1313, Jun. 2021.

[15] R. Liu, M. Li, H. Luo, Q. Liu, and A. L. Swindlehurst, "Integrated sensing and communication with reconfigurable intelligent surfaces: Opportunities, applications, and future directions," *IEEE Wireless Commun.*, vol. 30, no. 1, pp. 50–57, Feb. 2023.

[16] S. Li, B. Duo, M. D. Renzo, M. Tao, and X. Yuan, "Robust secure UAV communications with the aid of reconfigurable intelligent surfaces," *IEEE Wireless Commun.*, vol. 20, no. 10, pp. 6402–6417, Oct. 2021.

[17] H. Luo, R. Liu, M. Li, Y. Liu, and Q. Liu, "Joint beamforming design for RIS-assisted integrated sensing and communication systems," *IEEE Trans. Veh. Technol.*, vol. 71, no. 12, pp. 13393–13397, Dec. 2022.

[18] X. Song, D. Zhao, H. Hua, T. X. Han, X. Yang, and J. Xu, "Joint transmit and reflective beamforming for IRS-assisted integrated sensing and communication," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Apr. 2022, pp. 189–194.

[19] K. Zhi, C. Pan, H. Ren, K. K. Chai, and M. Elkashlan, "Active RIS versus passive RIS: Which is superior with the same power budget?," *IEEE Commun. Lett.*, vol. 26, no. 5, pp. 1150–1154, May 2022.

[20] Z. Zhang et al., "Active RIS vs. passive RIS: Which will prevail in 6G?," *IEEE Trans. Commun.*, vol. 71, no. 3, pp. 1707–1725, Mar. 2023.

[21] Z. Kang, C. You, and R. Zhang, "Active-IRS-aided wireless communication: Fundamentals, designs and open issues," *IEEE Wireless Commun.*, vol. 31, no. 3, pp. 368–374, Jun. 2024.

[22] Z. Peng, X. Liu, X. Liu, C. Pan, X. Chen, and H. Ren, "Performance analysis of active RIS-aided multi-pair full-duplex communications with spatial correlation and imperfect CSI," *Sci. China Inf. Sci.*, vol. 66, no. 9, Aug. 2023, Art. no. 192304.

[23] W. Lv, J. Bai, Q. Yan, and H. M. Wang, "RIS-assisted green secure communications: Active RIS or passive RIS?," *IEEE Wireless Commun. Lett.*, vol. 12, no. 2, pp. 237–241, Feb. 2023.

Authorized licensed use limited to: Xi'an Univ of Posts & Telecom. Downloaded on March 17,2026 at 01:10:00 UTC from IEEE Xplore. Restrictions apply.

[24] Y. Ma, M. Li, Y. Liu, Q. Wu, and Q. Liu, "Active reconfigurable intelligent surface for energy efficiency in MU-MISO systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 3, pp. 4103–4107, Mar. 2023.

[25] R. Long, Y. -C. Liang, Y. Pei, and E. G. Larsson, "Active reconfigurable intelligent surface-aided wireless communications," *IEEE Trans. Wireless Commun.*, vol. 20, no. 8, pp. 4962–4975, Aug. 2021.

[26] M. H. Khoshafa, T. M. N. Ngatched, M. H. Ahmed, and A. R. Ndjiongue, "Active reconfigurable intelligent surfaces-aided wireless communication system," *IEEE Commun. Lett.*, vol. 25, no. 11, pp. 3699–3703, Nov. 2021.

[27] L. Dong, H. -M. Wang, and J. Bai, "Active reconfigurable intelligent surface aided secure transmission," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 2181–2186, Feb. 2022.

[28] P. Zeng, D. Qiao, Q. Wu, and Y. Wu, "Throughput maximization for active intelligent reflecting surface-aided wireless powered communications," *IEEE Wireless Commun. Lett.*, vol. 11, no. 5, pp. 992–996, May 2022.

[29] Q. Zhu, M. Li, R. Liu, and Q. Liu, "Joint transceiver beamforming and reflecting design for active RIS-aided ISAC systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 7, pp. 9636–9640, Jul. 2023.

[30] A. A. Salem, M. H. Ismail, and A. S. Ibrahim, "Active reconfigurable intelligent surface-assisted MISO integrated sensing and communication systems for secure operation," *IEEE Trans. Veh. Technol.*, vol. 72, no. 4, pp. 4919–4931, Apr. 2023.

[31] K. -K. Wong, A. Shojaeifard, K. -F. Tong, and Y. Zhang, "Fluid antenna systems," *IEEE Wireless Commun.*, vol. 20, no. 3, pp. 1950–1962, Mar. 2021.

[32] L. Zhu, W. Ma, and R. Zhang, "Modeling and performance analysis for movable antenna enabled wireless communications," *IEEE Trans. Wireless Commun.*, vol. 23, no. 6, pp. 6234–6250, Jun. 2024.

[33] L. Zhu, W. Ma, and R. Zhang, "Movable antennas for wireless communication: Opportunities and challenges," *IEEE Commun. Mag.*, vol. 62, no. 6, pp. 114–120, Jun. 2024.

[34] W. K. New et al., "A tutorial on fluid antenna system for 6G networks: Encompassing communication theory, optimization methods and hardware designs," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 4, pp. 2325–2377, Aug. 2025.

[35] T. Wu et al., "Fluid antenna systems enabling 6G: Principles, applications, and research directions," 2024, *arXiv:2412.03839*.

[36] L. Zhu et al., "A tutorial on movable antennas for wireless networks," *IEEE Commun. Surveys Tuts.*, early access, Feb. 27, 2025, doi: 10.1109/COMST.2025.3546373.

[37] J. A. Zhang et al., "An overview of signal processing techniques for joint communication and radar sensing," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 6, pp. 1295–1315, Nov. 2021.

[38] J. Chu, R. Liu, M. Li, Y. Liu, and Q. Liu, "Joint secure transmit beamforming designs for integrated sensing and communication systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 4, pp. 4778–4791, Apr. 2023.

[39] D. N. Dao, H. Zhang, A. B. J. Kokkeler, and Y. Miao, "Joint beamforming for multi-user multi-target FD ISAC system: A hybrid GRQ-GA approach," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Mar. 2025, pp. 1–6.

[40] H. Hua, J. Xu, and T. X. Han, "Optimal transmit beamforming for integrated sensing and communication," *IEEE Trans. Veh. Technol.*, vol. 72, no. 8, pp. 10588–10603, Aug. 2023.

[41] F. Liu, L. Zhou, C. Masouros, A. Li, W. Luo, and A. Petropulu, "Toward dual-functional radar-communication systems: Optimal waveform design," *IEEE Trans. Signal Process.*, vol. 66, pp. 4264–4279, Aug. 2018.

[42] H. Zhang, "Joint waveform and phase shift design for RIS-assisted integrated sensing and communication based on mutual information," *IEEE Commun. Lett.*, vol. 26, no. 10, pp. 2317–2321, Oct. 2022.

[43] Y. Xu, Y. Li, J. A. Zhang, M. D. Renzo, and T. Q. S. Quek, "Joint beamforming for RIS-assisted integrated sensing and communication systems," *IEEE Trans. Commun.*, vol. 72, no. 4, pp. 2232–2246, Apr. 2024.

[44] Z. Yu et al., "Active RIS-aided ISAC systems: Beamforming design and performance analysis," *IEEE Trans. Commun.*, vol. 72, no. 3, pp. 1578–1595, Mar. 2024.

[45] M. Liu et al., "Joint beamforming design for double active RIS-assisted radar-communication coexistence systems," *IEEE Trans. Cogn. Commun. Netw.*, vol. 10, no. 5, pp. 1704–1717, Oct. 2024.

[46] Q. Zhu, M. Li, R. Liu, and Q. Liu, "Cramér-Rao bound optimization for active RIS-empowered ISAC systems," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 11723–11736, Sep. 2024.

[47] Z. Yang, S. Zhang, G. Chen, Z. Dong, Y. Wu, and D. B. da Costa, "Secure integrated sensing and communication systems assisted by active RIS," *IEEE Trans. Veh. Technol.*, vol. 73, no. 12, pp. 19791–19796, Dec. 2024.

[48] Y. Zhang et al., "Secure wireless communication in active RIS-assisted DFRC systems," *IEEE Trans. Veh. Technol.*, vol. 74, no. 1, pp. 626–640, Jan. 2025.

[49] H. Xu et al., "Capacity maximization for FAS-assisted multiple access channels," *IEEE Trans. Commun.*, vol. 73, no. 7, pp. 4713–4731, Jul. 2025, doi: 10.1109/TCOMM.2024.3516499.

[50] W. Ma, L. Zhu, and R. Zhang, "MIMO capacity characterization for movable antenna systems," *IEEE Trans. Wireless Commun.*, vol. 23, no. 4, pp. 3392–3407, Apr. 2024.

[51] Y. Gao, Q. Wu, and W. Chen, "Joint transmitter and receiver design for movable antenna enhanced multicast communications," *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 18186–18200, Dec. 2024.

[52] L. Zhu, W. Ma, B. Ning, and R. Zhang, "Movable-antenna enhanced multiuser communication via antenna position optimization," *IEEE Trans. Wireless Commun.*, vol. 23, no. 7, pp. 7214–7229, Jul. 2024.

[53] J. Tang, C. Pan, Y. Zhang, H. Ren, and K. Wang, "Secure MIMO communication relying on movable antennas," *IEEE Trans. Commun.*, vol. 73, no. 4, pp. 2159–2175, Apr. 2025.

[54] Z. Cheng, N. Li, J. Zhu, X. She, C. Ouyang, and P. Chen, "Sum-rate maximization for fluid antenna enabled multiuser communications," *IEEE Commun. Lett.*, vol. 28, no. 5, pp. 1206–1210, May 2024.

[55] Q. Zhang, M. Shao, T. Zhang, G. Chen, J. Liu, and P. C. Ching, "An efficient sum-rate maximization algorithm for fluid antenna-assisted ISAC system," *IEEE Commun. Lett.*, vol. 29, no. 1, pp. 200–204, Jan. 2025.

[56] W. Lyu, S. Yang, Y. Xiu, Z. Zhang, C. Assi, and C. Yuen, "Movable antenna enabled integrated sensing and communication," *IEEE Trans. Wireless Commun.*, vol. 24, no. 4, pp. 2862–2875, Apr. 2025.

[57] Y. Ye, L. You, H. Xu, A. Elzanaty, K. -K. Wong, and X. Gao, "SCNR maximization for MIMO ISAC assisted by fluid antenna system," *IEEE Trans. Veh. Technol.*, vol. 74, no. 8, pp. 13272–13277, Aug. 2025.

[58] C. Jiang, C. Zhang, C. Huang, J. Ge, D. Niyato, and C. Yuen, "Movable antenna-assisted integrated sensing and communication systems," *IEEE Trans. Wireless Commun.*, vol. 24, no. 8, pp. 6397–6412, Aug. 2025.

[59] H. Wu, H. Ren, C. Pan, and Y. Zhang, "Movable antenna-enabled RIS-aided integrated sensing and communication," *IEEE Trans. Cogn. Commun. Netw.*, vol. 11, no. 5, pp. 2879–2892, Oct. 2025.

[60] A. Zhuravlev, V. Razevig, S. Ivashov, A. Bugaev, and M. Chizh, "Experimental simulation of multi-static radar with a pair of separated movable antennas," in *Proc. IEEE Int. Conf. Microw., Commun., Antennas, Biomed. Eng. Electron. Syst.*, Nov. 2015, pp. 1–5.

[61] S. Basbug, "Design and synthesis of antenna array with movable elements along semicircular paths," *IEEE Antennas Wirel. Propag. Lett.*, vol. 16, pp. 3059–3062, 2017.

[62] Y. Wu, D. Xu, D. W. K. Ng, W. Gerstacker, and R. Schober, "Movable antenna-enhanced multiuser communication: Jointly optimal discrete antenna positioning and beamforming," in *Proc. IEEE Glob. Commun. Conf.*, Dec. 2023, pp. 7508–7513.

[63] J. Zou et al., "Shifting the ISAC trade-off with fluid antenna systems," *IEEE Wireless Commun. Lett.*, vol. 13, no. 12, pp. 3479–3483, Dec. 2024.

[64] W. Mei, X. Wei, B. Ning, Z. Chen, and R. Zhang, "Movable-antenna position optimization: A graph-based approach," *IEEE Wireless Commun. Lett.*, vol. 13, pp. 1853–1857, Jul. 2024.

[65] W. Mei, X. Wei, Y. Liu, B. Ning, and Z. Chen, "Movable-antenna position optimization for physical-layer security via discrete sampling," in *Proc. IEEE Glob. Commun. Conf.*, Dec. 2024, pp. 4750–4755.

[66] Z. Cheng, N. Li, J. Zhu, C. Ouyang, and X. Zhang, "Exploiting movable antennas in multicast communications," 2024, *arXiv:2405.07281*.

[67] M. Grant and S. Boyd, "CVX: MATLAB software for disciplined convex programming, version 2.1," Mar. 2014. [Online]. Available: http://cvxr.com/cvx

[68] X. Liu, T. Huang, N. Shlezinger, Y. Liu, J. Zhou, and Y. C. Eldar, "Joint transmit beamforming for multiuser MIMO communications and MIMO radar," *IEEE Trans. Signal Process.*, vol. 68, pp. 3929–3944, 2020.

[69] X.-D. Zhang, *Matrix Analysis and Applications*. Cambridge, U.K.: Cambridge Univ. Press, 2017.

[70] K. Y. Wang, A. M. C. So, T. H. Chang, W. K. Ma, and C. Y. Chi, "Outage constrained robust transmit optimization for multiuser MISO downlinks: Tractable approximations by conic optimization," *IEEE Trans. Signal Process.*, vol. 62, no. 21, pp. 5690–5705, Nov. 2014.

[71] S. Lu, F. Liu, and L. Hanzo, "The degrees-of-freedom in monostatic ISAC channels: NLoS exploitation vs. reduction," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 2643–2648, Feb. 2023.

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